Entanglement measures and LOCC conversion criteria from Schur-Weyl duality

Anton van Cleemput,[1,2] Otfried Gühne,[2] and Albert Rico[2]

[1] *Sorbonne Université, CNRS, LIP6, F-75005 Paris, France*

[2] *Naturwissenschaftlich-Technische Fakultät, Universität Siegen, Walter-Flex-Straße 3, 57068 Siegen, Germany*

(Dated: Annotated copy for the internship report, October 28, 2026)

The characterization of possible transformations between quantum states and the quantification of entanglement are central for understanding quantum resources. Explicit results are, however, scarce since they require complicated optimizations over transformation protocols or state decompositions. We introduce a method to optimize functions of the entanglement spectrum of bipartite states over mixed-state decompositions with arbitrary accuracy. Our results are based on a convergent semidefinite programming hierarchy of optimizations over entanglement spectra using a natural polynomial basis and Schur-Weyl duality. Our approach can also be used to determine non-polynomial measures, such as the entanglement of formation, and to characterize the dimensionality of entanglement. Moreover, extracting the decomposition allows us to formulate criteria to determine state transformations under local operations and classical communication (LOCC). In particular, we provide a hierarchy of certificates of LOCC non-convertibility from pure to mixed states in any dimension and a heuristic approach for LOCC convertibility.

How to read this copy. This is the arXiv version with numbered tags where the report should go further. **More detail**: explanations, examples, figures or numbers the paper leaves out for space. **Deeper proof**: proof steps that are sketched, only cited, or need checking. **New content**: material that is not in the paper (libraries, first phase, implementation). Put `\tagsfalse` in the preamble to hide all tags.

I. INTRODUCTION

Composite quantum systems are entangled when they cannot be described solely by their parts [1]. Operationally, entanglement is a resource for quantum communication [2, 3] and computation [4, 5] that cannot increase under local operations and classical communications (LOCC) between distant parties. In isolated two-particle quantum systems, described by pure states, entanglement and LOCC transformations are completely characterized by the Schmidt coefficients or the eigenvalues of the marginals [6, 7], also called the entanglement spectrum. Therefore, learning this spectrum is necessary and sufficient to understand the entanglement of the system and the possible entanglement transformations between states.

In practice, however, pure states become mixed due to decoherence with the environment [8–10]. When this occurs, the spectrum of the marginals alone is not informative of the entanglement between them, because a mixed state admits multiple descriptions as a probabilistic ensemble of pure states. Consequently, one needs to account for all such decompositions. This leads to an intractable optimization required to evaluate entanglement measures [11–14], even when full tomography is available. Existing methods address each specific entanglement measure independently [15–22]. Moreover, while lower bounds can be found for polynomial measures [23], much less is known for the more general case [24]. As a result, it is still a remarkable challenge to determine when entanglement transformations are possible with LOCC [25–27].

Here we aim to provide a systematic method to learn the entanglement properties of mixed bipartite quantum states. To this end we use the Schur-Weyl decomposition of the Hilbert space of multiple copies, which allows one to learn entanglement spectra of pure states [28, 29] to estimate any polynomial entanglement measure for mixed states with a complete polynomial basis. This basis is particularly suited to detect entanglement dimensionality. We find upper and lower converging bounds, which determine, for instance, the amount of entanglement of the main families of bound entangled states. Using semidefinite and linear programming, we extend them to a convergent hierarchy to find any possible entanglement measure of mixed states. The combination of these bounds leads us to a hierarchy of certificates to determine LOCC convertibility from pure to mixed and a heuristic approach towards convertibility.

[1] **More detail.** Write the introduction for a jury that is not made of specialists: (i) motivation with a concrete example (a Bell state used for teleportation or key distribution, and how noise makes it mixed); (ii) what quantifying entanglement means operationally (entanglement cost and distillation, of which E_F is the building block); (iii) a one-paragraph plain-language summary of the method, ideally with a pipeline figure: state → SDP → lower bound; SDP solution → extraction → upper bound; LP for non-polynomial measures; LOCC test. Close with the internship context (where, when, supervisors), the scope of what was produced (the paper, the `schur-weyl` and `bound-entangled` libraries, the first-phase Schmidt-number study, the code behind every figure) and an outline of the report.

II. ENTANGLEMENT SPECTRUM

Consider a bipartite pure state $|\psi\rangle_{AB} \in \mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B} \cong \mathbb{C}^D$ with $D = d_A d_B$. Such a state is separable if it can be written as $|\psi\rangle = |a\rangle \otimes |b\rangle$ and entangled otherwise. More generally, any pure bipartite state admits a *Schmidt decomposition*

$$|\psi\rangle_{AB} = \sum_{i=1}^{r} \sqrt{\alpha_i}\, |a_i\rangle \otimes |b_i\rangle\,, \tag{1}$$

where $\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_r > 0$ and the sets $\{|a_i\rangle\}_i$ and $\{|b_i\rangle\}_i$ are orthonormal. The spectrum $\alpha = (\alpha_1, \ldots, \alpha_{d_A})$ of the reduced density matrix $\rho_A = \mathrm{tr}_B\, |\psi\rangle\langle\psi|$ (with $\alpha_i = 0$ for $i > r$) is called the *entanglement spectrum* of $|\psi\rangle_{AB}$, and the number r of nonzero coefficients is known as the *Schmidt rank*, denoted $\mathrm{SR}(\psi)$.

[2] More detail. Derive the Schmidt decomposition from the singular value decomposition of the coefficient matrix ψ_{ij} (three lines), and introduce a running example: $\cos\theta\,|00\rangle + \sin\theta\,|11\rangle$, with spectrum $(\cos^2\theta, \sin^2\theta)$, from product ($\theta = 0$) to maximally entangled ($\theta = \pi/4$). Every measure below can be evaluated on it.

By the Nielsen theorem [6], a bipartite pure state $|\psi\rangle$ with entanglement spectrum $\alpha(\psi)$ can be converted by LOCC into another bipartite pure state $|\phi\rangle$ with entanglement spectrum $\alpha(\phi)$ if and only if $E_j(\psi) \geq E_j(\phi)$ for all $j = 1, \ldots, d_A - 1$, where

$$E_j(\psi) = 1 - \sum_{i=1}^{j} \alpha_i(\psi), \tag{2}$$

as defined in Ref. [30]. In this sense, the quantities $E_j(\psi)$ form a complete set of entanglement monotones, and all information about the entanglement of each state is encoded in the entanglement spectrum. Every pure-state entanglement monotone is a concave, unitarily invariant function of ρ_A [7].

[3] More detail. LOCC, entanglement monotones and majorization are used without definitions. Add: LOCC (with a picture of the two labs and the classical channel), entanglement monotones (non-increasing on average under LOCC), majorization with an illustration of Nielsen's theorem, and a table of the measures used later (linear entropy, entropy of entanglement, Vidal monotones E_j, Schmidt rank) evaluated on the running example.

[4] Deeper proof. Vidal's characterisation (a pure-state monotone is a concave, unitarily invariant function of ρ_A) is used several times; state it as a theorem and sketch the direction you need (concavity ⇒ monotonicity under Alice's local measurements).

A pure-state entanglement measure E can be extended to mixed states via the convex roof extension,

$$\widehat{E}(\rho) = \min_{\mathcal{D}(\rho)} \sum_i p_i E(\psi_i), \tag{3}$$

where the minimization is taken over all decompositions $\mathcal{D}(\rho) = \{p_i, |\psi_i\rangle\}$ of ρ such that $\rho = \sum_i p_i\, |\psi_i\rangle\langle\psi_i|$. However, this optimization is generally intractable, even if the state is known.

[5] More detail. Before the convex roof, define separable mixed states, the PPT (Peres–Horodecki) criterion, bound entanglement and the Schmidt number of a mixed state. All four are used later (the PPT constraints of the SDP, the bound entangled test states, the Schmidt-number result) but never defined. Then explain why the convex roof is hard: decompositions are parametrised by isometries (HJW, App. I 1), the problem is non-convex, the optimum may need up to r^2 terms, and closed forms exist only in special cases (Wootters' formula for two qubits, isotropic and Werner states). A figure of one two-qubit state written as two different ensembles helps here.

III. DECOMPOSITION OF THE HILBERT SPACE

Let us now explain facts about the Hilbert space of k copies of a state $|\psi\rangle$, since this is the basis of our approach. A general k-qudit quantum state is a vector in the Hilbert space $(\mathbb{C}^d)^{\otimes k}$. We consider two groups acting on this space: the *symmetric group* S_k of permutations $\pi \in S_k$ of k elements, acting as

$$\pi(|\psi_1\rangle \otimes \ldots \otimes |\psi_k\rangle) = \left|\psi_{\pi^{-1}(1)}\right\rangle \otimes \ldots \otimes \left|\psi_{\pi^{-1}(k)}\right\rangle, \tag{4}$$

and the *general linear group* GL_d of invertible $d \times d$ matrices $g \in GL_d$, acting as

$$g(|\psi_1\rangle \otimes \ldots \otimes |\psi_k\rangle) = g\,|\psi_1\rangle \otimes \ldots \otimes g\,|\psi_k\rangle\,. \tag{5}$$

These two actions commute,

$$\pi\, g = g\, \pi. \tag{6}$$

The so-called *Schur-Weyl duality* [31] states that under the action of these two groups, the Hilbert space $(\mathbb{C}^d)^{\otimes k}$ decomposes as

$$(\mathbb{C}^d)^{\otimes k} = \bigoplus_{\substack{\lambda \vdash k \\ \ell(\lambda) \leq d}} \mathcal{S}_k^\lambda \otimes \mathcal{U}_d^\lambda \tag{7}$$

into irreducible subspaces $\mathcal{S}_k^\lambda$ and $\mathcal{U}_d^\lambda$, which carry the irreducible representations of S_k and GL_d, respectively, indexed by partitions λ. The dimensions of $\mathcal{S}_k^\lambda$ and $\mathcal{U}_d^\lambda$ are denoted as f^λ and $s_\lambda(1, \ldots, 1)$, where s_λ is the so-called Schur polynomial (see also below). Each partition λ is represented as a Young diagram: a non-increasing vector of positive integers summing to k, whose length we denote by $\ell(\lambda)$. We write $\lambda \vdash k$ to denote all the partitions λ of k.

To each Young diagram λ we can associate a so-called *isotypic projector* Π^λ onto the subspace $\mathcal{S}_k^\lambda \otimes \mathcal{U}_d^\lambda$. This projector is given by

$$\Pi^\lambda = \frac{f^\lambda}{k!} \sum_{\pi \in S_k} \chi^\lambda(\pi)\, \pi, \tag{8}$$

where $\chi^\lambda(\pi)$ is the character of the cycle type of π in the irreducible representation λ. It follows that the isotypic projectors commute with both permutations and tensor powers of matrices,

$$\pi\, \Pi^\lambda = \Pi^\lambda\, \pi\,, \quad g\, \Pi^\lambda = \Pi^\lambda\, g\,. \tag{9}$$

[6] More detail. This is the densest section for a reader without representation theory. Add worked examples: (i) $k = 2$: $(\mathbb{C}^d)^{\otimes 2} = \mathrm{Sym}^2 \oplus \wedge^2$, $\lambda = (2), (1,1)$, $f^\lambda = 1$, dimensions $d(d \pm 1)/2$, $\Pi^{(2)} = (\mathbb{1} + \mathrm{SWAP})/2$; (ii) $k = 3$: draw the three Young diagrams, give the character table of S_3, and show that $\lambda = (2,1)$ has $f^\lambda = 2$, so it appears twice. Give the hook-length formula for f^λ and the hook-content formula for $s_\lambda(1, \ldots, 1)$, and explain why $\ell(\lambda) \leq d$ (antisymmetrising more than d vectors of $\mathbb{C}^d$ gives zero).

[7] Deeper proof. Derive the isotypic projector from the orthogonality of characters (it is the central idempotent of the group algebra) and show that the Π^λ are orthogonal projectors summing to the identity. Commutation with $g^{\otimes k}$ follows from Eq. (6) in one line; write it.

IV. POLYNOMIAL OPTIMIZATION OVER MIXED-STATE DECOMPOSITIONS

We are now ready to present the main tool that we use to address the problem of entanglement estimation and convertibility certification. The spectrum of a reduced state ρ_A can be estimated via weak Schur sampling on k copies of ρ_A [28, 29]: the probability of measuring a Young diagram $\lambda \vdash k$ is given by

$$\mathrm{tr}\big(\Pi^\lambda \rho_A^{\otimes k}\big) = f^\lambda s_\lambda(\mathrm{spec}\, \rho_A). \tag{10}$$

It is known that the Schur polynomials s_λ with $\lambda \vdash k$ and $\ell(\lambda) \leq d_A$ form a basis of the homogeneous symmetric polynomials P of degree k of the spectrum of ρ_A [32], and they expand into the monomial basis in the following way:

$$s_\lambda(\mathrm{spec}\, \rho_A) = \sum_{\mu \vdash k} K_{\lambda\mu} m_\mu(\mathrm{spec}\, \rho_A), \tag{11}$$

where m_μ is the monomial symmetric polynomial

$$m_\mu(\alpha) = \sum_\beta \alpha_1^{\beta_1} \alpha_2^{\beta_2} \ldots \alpha_{d_A}^{\beta_{d_A}} \tag{12}$$

where the sum runs over all *distinct* permutations β of $(\mu_1, \mu_2, \ldots, \mu_{\ell(\mu)}, 0, \ldots, 0) \in \mathbb{N}^{d_A}$, and $K_{\lambda\mu}$ is a constant known as the Kostka number [32].

[8] Deeper proof. Prove the weak Schur sampling formula $\operatorname{tr}\left(\Pi^\lambda \rho_A^{\otimes k}\right) = f^\lambda s_\lambda(\operatorname{spec}\rho_A)$: by unitary invariance take ρ_A diagonal; on $\mathcal{S}_k^\lambda \otimes \mathcal{U}_d^\lambda$ the operator $\rho_A^{\otimes k}$ acts as $\mathbb{1}_{f^\lambda} \otimes q_\lambda(\rho_A)$, and the character of the GL_d irrep at a diagonal matrix is the Schur polynomial. Check it by hand at $k = 2$: $\operatorname{tr}\left(\Pi^{(2)}\rho^{\otimes 2}\right) = (1 + \operatorname{tr}\rho^2)/2 = s_{(2)}$.

[9] More detail. Give a small Kostka example, e.g. $s_{(2,1)} = m_{(2,1)} + 2m_{(1,1,1)}$, and write the measures you use in the Schur basis. The linear entropy is the cleanest: at $k = 2$, $1 - \operatorname{tr}\rho_A^2 = 2s_{(1,1)}(\operatorname{spec}\rho_A)$, so $O_P = 2\Pi^{(1,1)}$, twice the antisymmetric projector. Then say explicitly that E_F and the Vidal monotones are *not* polynomials, which motivates Sec. VIII.

Since $\operatorname{tr}(\rho_A) = 1$ implies $\operatorname{tr}(\rho_A)^l = 1$, any polynomial of degree $l < k$ can be lifted to a homogeneous polynomial of degree k. Thus, any symmetric polynomial of degree at most k of the entanglement spectrum can be expressed as

$$P(\operatorname{spec}\rho_A) = \sum_{\lambda \vdash k} c_\lambda s_\lambda(\operatorname{spec}\rho_A). \tag{13}$$

In order to estimate any entanglement measure of bipartite states given by a symmetric polynomial, we are interested in the convex roof of these polynomials,

$$\widehat{P}(\rho_{AB}) = \min_{\mathcal{D}(\rho_{AB})} \sum_i p_i \operatorname{tr}\left(O_P\, \rho_{A,i}^{\otimes k}\right), \tag{14}$$

where $\rho_{A,i} = \operatorname{tr}_B |\psi_i\rangle\langle\psi_i|$ and

$$O_P = \sum_{\lambda \vdash k} \frac{c_\lambda}{f^\lambda} \Pi^\lambda \tag{15}$$

for any tuple of coefficients $\vec{c} := (c_{\lambda_1}, \ldots, c_{\lambda_t})$, where t is the number of partitions of k. $\widehat{P}$ is an entanglement monotone if P is concave [7]. For any P, we always have unitary invariance since P only depends on the spectrum.

[10] More detail. Spell out the identity the whole method rests on, which is only implicit here: $P(\operatorname{spec}\rho_{A,i}) = \operatorname{tr}\left(O_P\, \rho_{A,i}^{\otimes k}\right) = \operatorname{tr}\left((O_P \otimes \mathbb{1}^B)\, |\psi_i\rangle\langle\psi_i|^{\otimes k}\right)$, so $\sum_i p_i P(\psi_i) = \operatorname{tr}\left((O_P \otimes \mathbb{1}^B)\, \omega_k\right)$ is *linear* in the operator ω_k of Eq. (16). With this computation in front, the next section reads naturally.

V. A COMPLETE HIERARCHY FOR ENTANGLEMENT ESTIMATION

Our approach consists of using the spectrum estimation formulation above for mixed-state decompositions. To this end, we use that polynomial entanglement measures can be bounded using optimizations over functions of several copies of a state [23]. That is, given a bipartite mixed state ρ_{AB} acting on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$, to any decomposition in $\mathcal{D}(\rho_{AB})$ we associate the state

$$\omega_k = \sum_i p_i\, |\psi_i\rangle\langle\psi_i|^{\otimes k} \tag{16}$$

acting on $(\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B})^{\otimes k}$. The core idea is that a polynomial entanglement measure of degree not larger than k can be written as a linear function on ω_k. Then, optimizing over all decompositions amounts to optimizing over all ω_k with the constraint that ω_k is symmetric and fully separable with respect to the k copies. The separability constraint is difficult to implement, but it can be treated by a hierarchy of semidefinite approximations, similar to the separability criterion of symmetric extensions [33].

[11] More detail. Add background on semidefinite programming: primal and dual forms, weak and strong duality, Slater's condition, interior-point methods, and what "up to solver precision" means for a certificate. Present the DPS hierarchy of symmetric extensions [33] as the template, and Ref. [23] as the starting point: what it did (linear entropy, Schmidt number, three-tangle) and what is new here. The list from the draft's commented-out block is a good skeleton: any symmetric polynomial via the Schur basis and isotypic projectors; convergence via de Finetti; reduction to the symmetric subspace of the support and to real variables; the cheaper simplified hierarchy; the embedded LP for non-polynomial measures and LOCC; extraction of decompositions for upper bounds.

Explicitly, we can compute a sequence of lower bounds on Eq. (14) as $\widehat{P}(\rho) \geq P_{\mathrm{LB}_k}(\rho)$ with the following semidefinite programming (SDP) relaxation,

$$
\begin{aligned}
P_{\mathrm{LB}_k}(\rho) = \min_{\Omega_k} \ & \operatorname{tr}\left(O_P^A \otimes \mathbb{1}^B \, \Omega_k\right) \\
\text{s.t. } & \Omega_k \succeq 0, \\
& \Pi_{\mathrm{sym}}^{AB,k} \, \Omega_k = \Omega_k \, \Pi_{\mathrm{sym}}^{AB,k} = \Omega_k, \\
& \operatorname{tr}_{2,\ldots,k}(\Omega_k) = \rho, \\
& \Omega_k^{T_l} \succeq 0 \quad \forall l \in \{1, \ldots, \lfloor \tfrac{k}{2} \rfloor\}.
\end{aligned}
\tag{17}
$$

Here, O_P^A acts on $A_1, \ldots, A_k$. When $\deg(P) = m < k$, P can be lifted to degree k, or, equivalently, evaluated on $A_1, \ldots, A_m$ as both give the same bound (Lemma 8). $\Pi_{\mathrm{sym}}^{AB,k}$ is the projector onto the k-copy symmetric subspace of $(\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B})^{\otimes k}$ across copies, and T_l denotes the partial transpose on the first l copies. Since every ω_k is in the feasible set of this SDP at every level k, it gives a lower bound on the convex roof. This yields an SDP hierarchy of k-th order lower bounds. This hierarchy is complete, as shown in Appendix A.

[12] More detail. Check each constraint on the true ω_k: positivity; Bose symmetry ($|\psi_i\rangle^{\otimes k} \in \mathrm{Sym}^k$); the marginal ($\operatorname{tr}_{2,\ldots,k} \omega_k = \sum_i p_i \, |\psi_i\rangle\langle\psi_i| = \rho$); and PPT across copies (each term is a product across copies, so every partial transpose is again a mixture of positive operators). This shows that every ω_k is feasible, which is the whole reason the SDP gives a lower bound.

Result 1. *The bound of Eq. (17) is saturated asymptotically,* $\lim_{k\to\infty} P_{\mathrm{LB}_k}(\rho) = \widehat{P}(\rho)$.

The proof sketch is as follows: by the de Finetti theorem for Bose-symmetric states [34, 35], the m-copy marginal of any feasible Ω_k is asymptotically close to a mixture $\sum p_i \, |\phi_i\rangle\langle\phi_i|^{\otimes m}$ whose first system marginal is asymptotically close to ρ as k increases. By continuity of the convex roof, $P_{\mathrm{LB}_k} \geq \widehat{P}(\rho) - \delta_k$ where $\lim_{k\to\infty} \delta_k = 0$. Moreover, the bound is nondecreasing in k (Appendix C).

[13] More detail. Show the convergence in practice: P_{LB_k} against k for one or two states, together with the upper bound, so the reader sees how fast the gap closes and at which level memory stops you.

Since Ω_k is supported on the symmetric subspace $\mathrm{Sym}^k(\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}) \subset (\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B})^{\otimes k}$, we can exploit this symmetry to reduce the matrix size of Ω_k from D^k to $\dim \mathrm{Sym}^k(\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}) = \binom{k+D-1}{k}$. By projecting onto the range of ρ, we can further reduce this size to $\binom{k+r-1}{k}$, with $r = \operatorname{rank} \rho$. Moreover, the variable Ω_k can be taken real for real states. These symmetry reductions are explained in detail in Appendix D.

[14] More detail. Make the savings concrete with a table of D^k, $\binom{k+D-1}{k}$ and $\binom{k+r-1}{k}$ for the states in the figures (e.g. two qutrits, $D = 9$, $k = 3$: $729 \to 165$, and 20 for $r = 4$), with the resulting memory and run time.

Naturally, using the isotypic polynomial basis allows us to detect the entanglement dimension, i.e. the Schmidt number [36], of a target mixed state. To see this, recall that $\Pi^\lambda \, |e_1\rangle \otimes \cdots \otimes |e_k\rangle = 0$ if fewer than $\ell(\lambda)$ distinct vectors of an orthonormal basis $\{|e_i\rangle\}$ appear in the product, and therefore $\operatorname{tr}\left(\Pi_A^\lambda \otimes \mathbb{1}_B \, |\psi\rangle\langle\psi|_{AB}^{\otimes k}\right) = 0$ if and only if $\ell(\lambda) > \mathrm{SR}(\psi_{AB})$ [37, 38]. Therefore, if $P_{\mathrm{LB}_k} \neq 0$ up to the solver precision, then no pure-state decomposition of ρ exists in which all states have Schmidt rank lower than $\ell(\lambda)$. This is formally described as follows:

Result 2. *If* $P_{\mathrm{LB}_k} > 0$ *is certified for* $O_P = \Pi^\lambda$ *with* $\lambda \vdash k$, *then* ρ *has Schmidt number at least* $\ell(\lambda)$.

As a simple example, we test this Schmidt number hierarchy on the state $\rho = p \, |\phi^+\rangle\langle\phi^+| + (1-p) \, |01\rangle\langle 01|$ acting on $\mathbb{C}^3 \otimes \mathbb{C}^3$, with $|\phi^+\rangle = \frac{1}{\sqrt{3}} \sum_{i=0}^{2} |ii\rangle$, studied in Ref. [39], which has Schmidt number 3 for $p > 0$, and we detect it from $p \approx 0.01$ at level $k = 3$ with $\lambda = (1,1,1)$. It is easy to see that the state ρ has Schmidt number three for any $p > 0$, since every pure state in a decomposition lies in the span of $|\phi^+\rangle$ and $|01\rangle$, and any such state has Schmidt rank three.

[15] Deeper proof. Prove the Schmidt-number property rather than citing it: if $|\psi\rangle$ has Schmidt rank r, then $\rho_A^{\otimes k}$ is supported on $(\mathbb{C}^r)^{\otimes k}$, where $\Pi^\lambda = 0$ for $\ell(\lambda) > r$; conversely, for $\ell(\lambda) \leq r$, $s_\lambda(\alpha) > 0$, since $s_\lambda = \sum_\mu K_{\lambda\mu} m_\mu$ with $K_{\lambda\mu} \geq 0$, $K_{\lambda\lambda} = 1$, and $m_\lambda(\alpha) > 0$ when r entries of α are positive. Also tighten the example: not every state in the span has Schmidt rank three ($|01\rangle$ does not), but every decomposition contains a state with a nonzero $|\phi^+\rangle$ component, and those have Schmidt rank three.

[16] New content. The first phase of the internship (May–September, `QuantifyingSchmidtNumber`) fits here: the Schmidt-number and E_t bounds, the state sweeps, the upper bounds from the September extraction (the `HANDOVER.md` table, e.g. isotropic $F = 0.8$ tight at $\lambda = (1,1,1)$ and $(1,1)$), and how they compare with this witness. A section with these results shows the path that led to the paper.

VI. SIMPLIFIED RELAXATION

Notice that the isotypic projectors Π^λ only act on Alice's subsystem copies. This leads us to ask whether the SDP can be simplified. In turn, here we present a simplification of the hierarchy above using a smaller variable that allows us to access larger systems, at the cost of weaker bounds at each level $P_{\mathrm{LB}_{k,m}} \leq P_{\mathrm{LB}_m}$ with $P_{\mathrm{LB}_{k,m}}$ defined in Eq. (19) below. For $m \geq k$, define the following extension acting on $(\mathbb{C}^{d_A})^{\otimes m} \otimes (\mathbb{C}^{d_B})^{\otimes k}$ (i.e., m copies of A and k copies of B),

$$\omega_{k,m} := \sum_i p_i \, |\psi_i\rangle\langle\psi_i|^{\otimes k} \otimes \rho_{A,i}^{\otimes(m-k)} \tag{18}$$

with $\rho_{A,i} = \mathrm{tr}_B(|\psi_i\rangle\langle\psi_i|)$. We relax this operator with a positive semidefinite variable $\Phi_{k,m}$, and denote $\Phi_k := \mathrm{tr}_{A_{k+1},\ldots,A_m}(\Phi_{k,m})$. Then we can lower bound $\widehat{P}(\rho)$ as $P_{\mathrm{LB}_{k,m}} \leq \widehat{P}(\rho)$, where

$$\begin{aligned} P_{\mathrm{LB}_{k,m}} := \min_{\Phi_{k,m}} \ & \mathrm{tr}\big(O_P^A \otimes \mathbb{1}^B \ \Phi_{k,m}\big) \\ \text{s.t.} \ & \Phi_{k,m} \succeq 0, \\ & \Pi_{\mathrm{sym}}^{AB,k} \, \Phi_k \, \Pi_{\mathrm{sym}}^{AB,k} = \Phi_k, \\ & \mathrm{tr}_{A_2,\ldots,A_m,B_2,\ldots,B_k}(\Phi_{k,m}) = \rho_{AB}, \\ & \mathrm{tr}_{B_1,\ldots,B_k}(\Phi_{k,m}) \text{ is permutation invariant}, \\ & \Phi_{k,m}^{T_S} \succeq 0 \quad \forall\, S. \end{aligned} \tag{19}$$

Here O_P^A is the observable of Eq. (15) for a polynomial of degree m, acting on $A_1, \ldots, A_m$, and T_S denotes the partial transpose on a set S of registers, where transposing a full copy acts on both A_i and B_i. Since $\Phi_{k,m}$ can be assumed invariant under permutations of $A_1B_1, \ldots, A_kB_k$ and $A_{k+1}, \ldots, A_m$, it suffices to impose partial transpose conditions on the tuples (i, j) where each tuple means taking the partial transpose on the first i full-copy registers and the first j marginal registers. One such set of representatives is $\{(i,j) : 0 \leq i \leq k,\ 0 \leq j < \frac{m-k}{2}\} \cup \{(i, \frac{m-k}{2}) : 0 \leq i \leq \lfloor k/2 \rfloor\}$ where the second set is only present for even $m-k$ and the tuple $(0,0)$ is excluded. For $k \geq \deg P$, the marginal Φ_k of any feasible $\Phi_{k,m}$ is feasible for Eq. (17) at level k, so that $P_{\mathrm{LB}_k} \leq P_{\mathrm{LB}_{k,m}}$ and this hierarchy is also complete in k. In fact, notice that for $k = m$ the hierarchy of Eq. (19) reduces to Eq. (17).

[17] More detail. Explain the idea behind $\omega_{k,m}$: the objective only needs many copies of the marginals $\rho_{A,i}$, not of the full $|\psi_i\rangle$, so k full copies tie the variable to ρ and $m-k$ cheap marginal copies raise the degree. Check each constraint on $\omega_{k,m}$ as for the full SDP, and give the counting argument behind the set of PPT representatives, with a small case such as $k = 2$, $m = 4$.

[18] Deeper proof. Write out $P_{\mathrm{LB}_k} \leq P_{\mathrm{LB}_{k,m}}$. It needs the lifting lemma on the A registers only, using that $\mathrm{tr}_B \, \Phi_{k,m}$ is permutation invariant (the fourth constraint), to move O_P from $A_1, \ldots, A_m$ to $A_1, \ldots, A_k$; otherwise the objective is not defined on Φ_k. Also prove the claim at the start of the section, $P_{\mathrm{LB}_{k,m}} \leq P_{\mathrm{LB}_m}$: tracing $B_{k+1}, \ldots, B_m$ out of a feasible Ω_m gives a feasible $\Phi_{k,m}$ with the same objective.

The parameters m and k can be used for a tradeoff between performance and efficiency. In terms of memory usage, this hierarchy is not necessarily less costly at level (k_1, m_1) with $k_1 < m_1$ than it is at level (k_2, m_2) with $k_2 = m_2$. This depends on the rank of the target state ρ and its marginals (See App. D 5). Typically, level (k_1, m_1) is less costly for full-rank states while (k_2, m_2) is less costly for low-rank ones.

Observation 3. *For full-rank states, the simplified hierarchy achieves tighter bounds at the same computational cost.*

We show the memory cost of the hierarchy at different levels for states of different ranks in Fig. 9 in the Appendix.

[19] More detail. The observation has no evidence in the main text. Give the memory model (about $8n^2$ bytes, where the number n of free variables depends only on r, $r_A = \mathrm{rank}\,\rho_A$ and the level), its rule of thumb (the full hierarchy is cheaper when $r < (k+1)\, r_A - k$), move Fig. 9 here, and add a table comparing the two hierarchies at equal memory. The story of the memory measurement that wrongly included the Python model assembly (3.2 GB against 0.1 GB in the solver) belongs in the implementation chapter.

VII. APPROXIMATING THE OPTIMAL DECOMPOSITION

Our next goal is to extract the decomposition $\{p_i, |\psi_i\rangle\}$ from the optimal solution Ω_k found in the SDP. This will provide us control of how tight our estimates are. Notice that if the relaxation is tight, we have $\Omega_k = \omega_k$. This is in

general not the case, and therefore we write $\Omega_k = \omega_k + \Delta$, where Δ accounts for the gap of the relaxation and solver noise. For $k \geq 3$, this decomposition is unique up to phases when the operators $|\psi_i\rangle\langle\psi_i|$ are linearly independent [40]. When the relaxation is tight and $\Omega_k = \omega_k$, we can then retrieve it exactly with Jennrich's algorithm [41, 42]. Since Ω_k is in general not of this form, we instead use the PARAFAC (or CANDECOMP) algorithm [41, 43] from the TensorLy package [44], initialized with Jennrich's algorithm, to find a decomposition $\{p_j, |\psi_j\rangle\}$ that locally minimizes the Frobenius norm

$$\Theta_k := \left\| \Omega_k - \sum_j p_j \, |\psi_j\rangle\langle\psi_j|^{\otimes k} \right\|_F, \tag{20}$$

with $\|X\|_F = \sqrt{\sum_{i,j} |X_{ij}|^2}$. Details are given in Appendix E 1.

[20] More detail. Tensor decompositions are new to most physics readers. Add background: CP rank; why matrix decompositions are not unique but order-3 tensor decompositions are (Kruskal's condition with k-ranks); Jennrich's algorithm step by step (two random contractions, simultaneous diagonalisation, read off the factors); alternating least squares for PARAFAC. A toy example (two qubits, a three-term decomposition recovered exactly from ω_3) shows the method working before the noisy case.

We obtain a decomposition of the state $\sigma = \sum_j p_j \, |\psi_j\rangle\langle\psi_j|$, which is close to, but generally different from ρ. We turn it into an exact decomposition of ρ by defining

$$|\varphi_j\rangle = \rho^{1/2}\sigma^{-1/2}\sqrt{p_j}\,|\psi_j\rangle\,, \tag{21}$$

where $\sigma^{-1/2}$ is taken on the support of σ. If $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$, then $\sum_j |\varphi_j\rangle\langle\varphi_j| = \rho^{1/2}\Pi_\sigma\rho^{1/2} = \rho$, with Π_σ the projector onto the support of σ. Hence $\{q_j, |\phi_j\rangle\}$ with $q_j = \|\varphi_j\|^2$ and $|\phi_j\rangle = |\varphi_j\rangle / \|\varphi_j\|$ is a decomposition of ρ and gives a valid upper bound on the convex roof.

[21] Deeper proof. Prove the repair identity $\sum_j |\varphi_j\rangle\langle\varphi_j| = \rho^{1/2}\Pi_\sigma\rho^{1/2} = \rho$ in two lines, and state the continuity property that the convergence proof needs: the repaired value approaches the unrepaired one as $\sigma \to \rho$. This is clear when $\operatorname{supp}\sigma = \operatorname{supp}\rho$, but $\sigma^{-1/2}$ blows up on small eigenvalues of σ outside $\operatorname{supp}\rho$, so check that case. Say what happens when $\operatorname{supp}\rho \not\subseteq \operatorname{supp}\sigma$ (too few terms).

As a comparison, we also compute upper bounds by gradient descent over ensemble decompositions, which can be unitarily parametrized. This result was first proven by Schrödinger [45], but regularly rediscovered [46–49] (App. I 1). Both upper bounds are comparable for the linear entropy and the Vidal monotone E_1, while gradient descent gives tighter upper bounds for the entanglement of formation (Figs. 5, 6 and 7). However, the tensor extraction method remains useful in cases where an upper bound cannot be extracted otherwise. The main result concerning optimal decompositions can be summarized as follows.

[22] More detail. State the Schrödinger–HJW theorem with its short proof (two decompositions of ρ are related by an isometry), then describe the gradient descent: parametrisation of the unitary, optimiser, number of restarts, why $M = r^2$, run times. The comparison of extraction and gradient descent is a result of its own: show where each wins and explain why extraction loses for E_F (the SDP optimises a polynomial lower bound, not S itself). The negative result that TensorLy's PARAFAC only converged with Jennrich initialisation belongs here too.

Result 4. *Let P be a symmetric polynomial in pure-state Schmidt coefficients. It is possible to extract the decomposition of Ω_k optimizing P. By doing so, we obtain an upper bound $P_{\mathrm{UB}_k}(\rho) \geq \widehat{P}(\rho)$ which converges,* $\lim_{k\to\infty} P_{\mathrm{UB}_k}(\rho) = \widehat{P}(\rho)$.

The proof is shown in App. E 2, and for it we use the rank reduction step (App. E 1), an instance of Jennrich's algorithm [41, 42] and the repair step from Eq. (21). In practice, we cannot run the SDP at k large enough to use Jennrich's algorithm and the extracted decomposition is always a heuristic one from PARAFAC.

[23] More detail. Report how well the heuristic works: the fit error Θ_k; the number of terms found against r and r^2; how much the repair step changes the value; and the effect of rank reduction (for two qubits the residual dropped from about 20 % to 10^{-7}–10^{-11}). The rank observation from `material.md` is worth a paragraph: for the tight states the rank of Ω_3 equals the number of terms (tiles 5, pyramid 5, chessboard 10, Horodecki 2×4 7), but rank flatness alone is not sufficient.

We show the bounds on the convex roof of the linear entropy $S_L(\psi) = 1 - \operatorname{tr}\rho_A^2$ in Fig. 1.

VIII. NON-POLYNOMIAL MEASURES VIA LINEAR PROGRAMMING

So far we have introduced a method to estimate polynomial entanglement measures, by using a natural basis of symmetric polynomials in the entanglement spectrum. Here we aim to use this approach to estimate non-polynomial

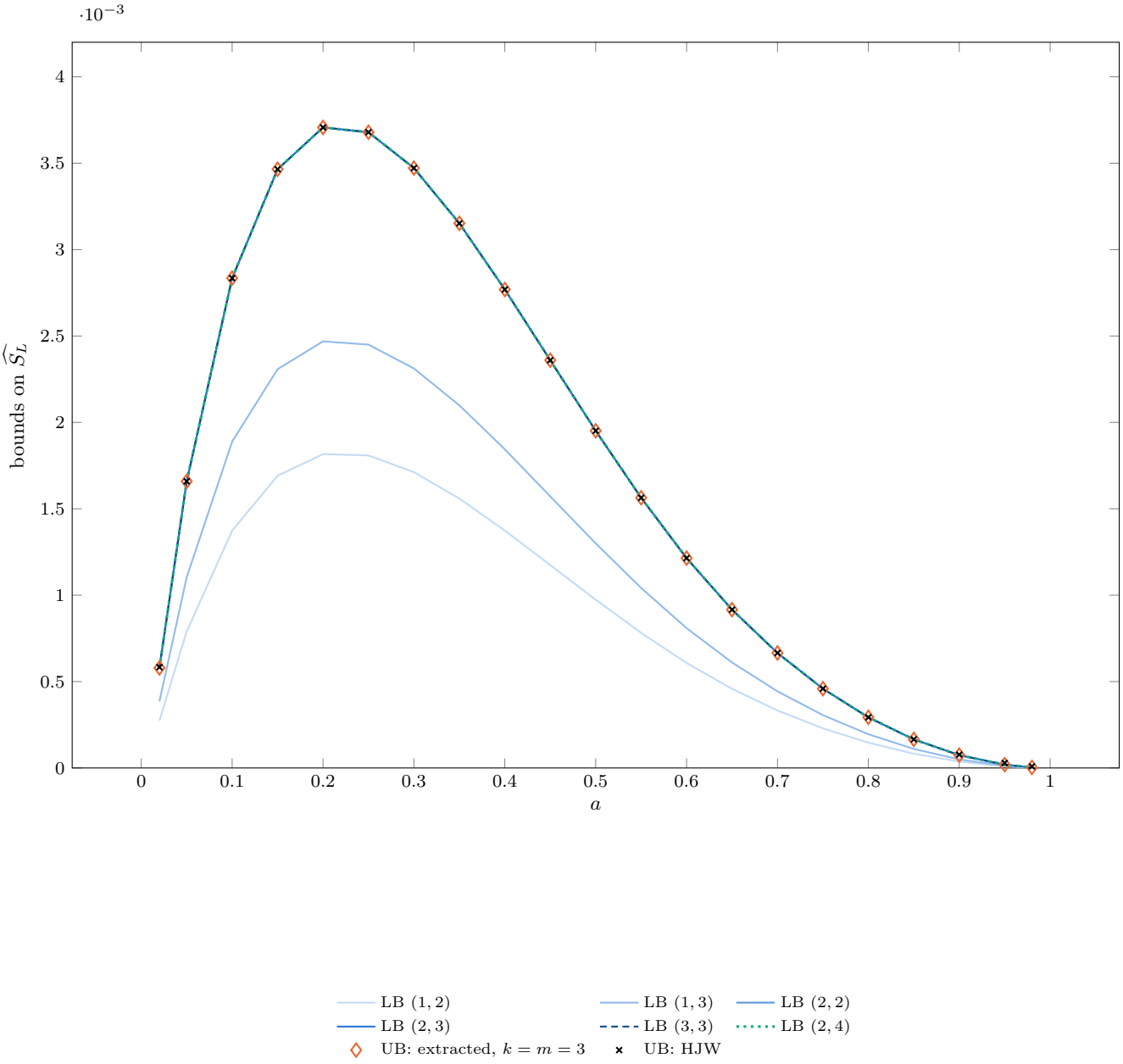


FIG. 1. Convex roof of the linear entropy $S_L = 1 - \operatorname{tr}\rho_A^2$ for the Horodecki 2×4 family. Lower bounds are shown at different levels of the hierarchy. The markers show the upper bounds extracted from our method and HJW gradient descent.

entanglement measures. To this end notice that all continuous entanglement measures can be bounded arbitrarily well by infinite series of polynomials. Our goal is therefore to find the polynomial that best approximates the target measure. Namely, we aim to find

$$\widetilde{F}_{\mathrm{LB}_k}(\rho) = \sup_{\substack{P \leq F \\ \deg P \leq k}} \min_{\Omega_k} \operatorname{tr}\left((O_P^A \otimes \mathbb{1}^B)\, \Omega_k \right), \tag{22}$$

the optimization over symmetric polynomials of degree at most k that lower bound F on all spectra.

Using that all such polynomials can be written in the (discrete) basis of Schur polynomials, we will show that finding the best polynomial that lower bounds any entanglement measure is a problem that can be cast as a linear program (LP) over the coefficients in this basis. This allows us to dynamically find the best lower-bounding polynomial for a given state while running the SDP.

The proof is constructive, namely we will present how to construct such LP. First, we select multiple spectra

$\alpha = (\alpha_1, \ldots, \alpha_{d_A})$ at which we compute all the Schur polynomials $s_\lambda(\alpha)$ of degree k and the input function $F(\alpha)$. These are points of the simplex of possible spectra α, and define a grid $G = \{\alpha^g\}$ discretizing this simplex of possible spectra (Fig. 3). Here the superindex α^g is chosen to distinguish the grid point g from the subindex j denoting each corresponding eigenvalue α_j^g. We then require $F(\alpha^g) \geq \sum_\lambda c_\lambda s_\lambda(\alpha^g)$ at every spectrum $\alpha^g \in G$. By symmetry of the Schur polynomials, we only need to compute the points where $\alpha_i \geq \alpha_j$ for $i < j$ for each spectrum α. Thus, we are left with the following optimization problem,

$$F_{\mathrm{LB}_k}(\rho) = \max_{\substack{P(\alpha^g) \leq F(\alpha^g)\, \forall g \\ \deg P \leq k}} \; \min_{\Omega_k} \; \mathrm{tr}\left((O_P^A \otimes \mathbb{1}^B)\, \Omega_k \right). \tag{23}$$

In order to solve this optimization, we need to address two challenges. First, a certain error ϵ_k is inherent to this grid procedure, in the sense that $|F_{\mathrm{LB}_k} - \widetilde{F}_{\mathrm{LB}_k}| \leq \epsilon_k$ (Fig. 3). Since we can retrieve the polynomial as output from the SDP, we can compute this error numerically and subtract it from the output to obtain a lower bound, namely $\widetilde{F}_{\mathrm{LB}_k} \geq F_{\mathrm{LB}_k} - \epsilon_k$. Second, this is in principle a min-max optimization problem. However, since the LP is feasible and has a bounded solution, strong duality holds, and we can turn this problem into a single minimization problem by taking the dual of the LP. Details are given in App. F.

[24] More detail. Add LP background (primal–dual pair, strong duality) and state Sion's minimax theorem with its hypotheses, since the derivation uses both. Describe the grid concretely: spacing, number of ordered points as a function of d_A and spacing, and how it scales. The work on which polynomial class to optimise over is not in the paper and fits here: the three classes (A: $g \geq 0$ with exact zeros, B: zeros only, C: only $g \leq f$), with the Werner-state example where C is exact (0.142929 at degree 2) and A gives 0.1225, and C converging in 3–12 cutting-plane rounds against about 50 for A and B. Also the comparison with the Taylor cubic and the Schur-sampling estimator $\sum_\lambda f(\lambda/k)\Pi^\lambda$.

[25] Deeper proof. Two statements are used without proof. (i) $|F_{\mathrm{LB}_k} - \widetilde{F}_{\mathrm{LB}_k}| \leq \epsilon_k$: one direction is $F_{\mathrm{LB}_k} \geq \widetilde{F}_{\mathrm{LB}_k}$ (more admissible polynomials), the other is that $P^* - \epsilon$ is admissible for $\widetilde{F}_{\mathrm{LB}_k}$. (ii) The grid error $\epsilon = \max_\alpha(P^* - F)$ is itself a non-convex maximisation. Say how you compute it. For the LOCC runs it was checked on a 1/600 lattice, which is an estimate rather than a certificate; a rigorous version needs e.g. a Lipschitz bound on $P^* - F$ times the lattice spacing (careful: F is not Lipschitz at the boundary for the entropy).

Result 5. *We obtain a complete hierarchy to estimate any entanglement measure on mixed states, in the following sense:*

1. *Let F be a continuous symmetric function of the entanglement spectrum, and let $\widehat{F}(\rho)$ be its convex roof (3). Then $\widetilde{F}_{\mathrm{LB}_k}(\rho) \leq \widehat{F}(\rho)$ at all levels k.*
2. *The bound is non-decreasing in k and convergent, $\lim_{k\to\infty} \widetilde{F}_{\mathrm{LB}_k}(\rho) = \widehat{F}(\rho)$.*
3. *With a finite grid G of spectra $\alpha^g \in G$, $F_{\mathrm{LB}_k}(\rho) - \epsilon \leq \widehat{F}(\rho)$, where $\epsilon = \max_{\alpha^g}\left(P^*(\alpha^g) - F(\alpha^g)\right)$ is the grid error of the polynomial P^* returned by the solver.*

The proofs of all statements above are given in Appendix F 2. We apply this method to estimate the convex roof of the von Neumann entropy of entanglement,

$$S(|\psi_{AB}\rangle) = -\,\mathrm{tr}(\rho_A \log_2 \rho_A), \tag{24}$$

where $\rho_A = \mathrm{tr}_B\, |\psi_{AB}\rangle\langle\psi_{AB}|$, extended to mixed states via convex roof, which we denote as $E_F = \widehat{S}$. This is known as the entanglement of formation [11], and it is well known to be difficult to compute [20]. To benchmark the method introduced, we use it to estimate $\widehat{S}(\rho_{AB})$ for several families of mixed states ρ_{AB} of different local dimensions, including the main known bound entangled states. The results are given in Fig. 2, and in more detail in Fig. 5 in the Appendix.

[26] More detail. Discuss the E_F results state by state: which states and why, why bound entangled states have small E_F, how close the lower and upper bounds get (e.g. only LB/UB ≈ 0.5 for 3×3 bound entangled states at $m = 3$, against ≈ 0.98 for E_1), the exact two-qubit values as a sanity check, and run times per level. A table of the numbers behind Fig. 2 makes them citable.

IX. LOCC CONVERTIBILITY

Finally, let us demonstrate how the presented approach allows us to determine whether certain pairs of states can be transformed by means of LOCC. To date, and to the best of our knowledge, the only criteria known for this problem are restricted to qubits and very special families of states [25, 26].

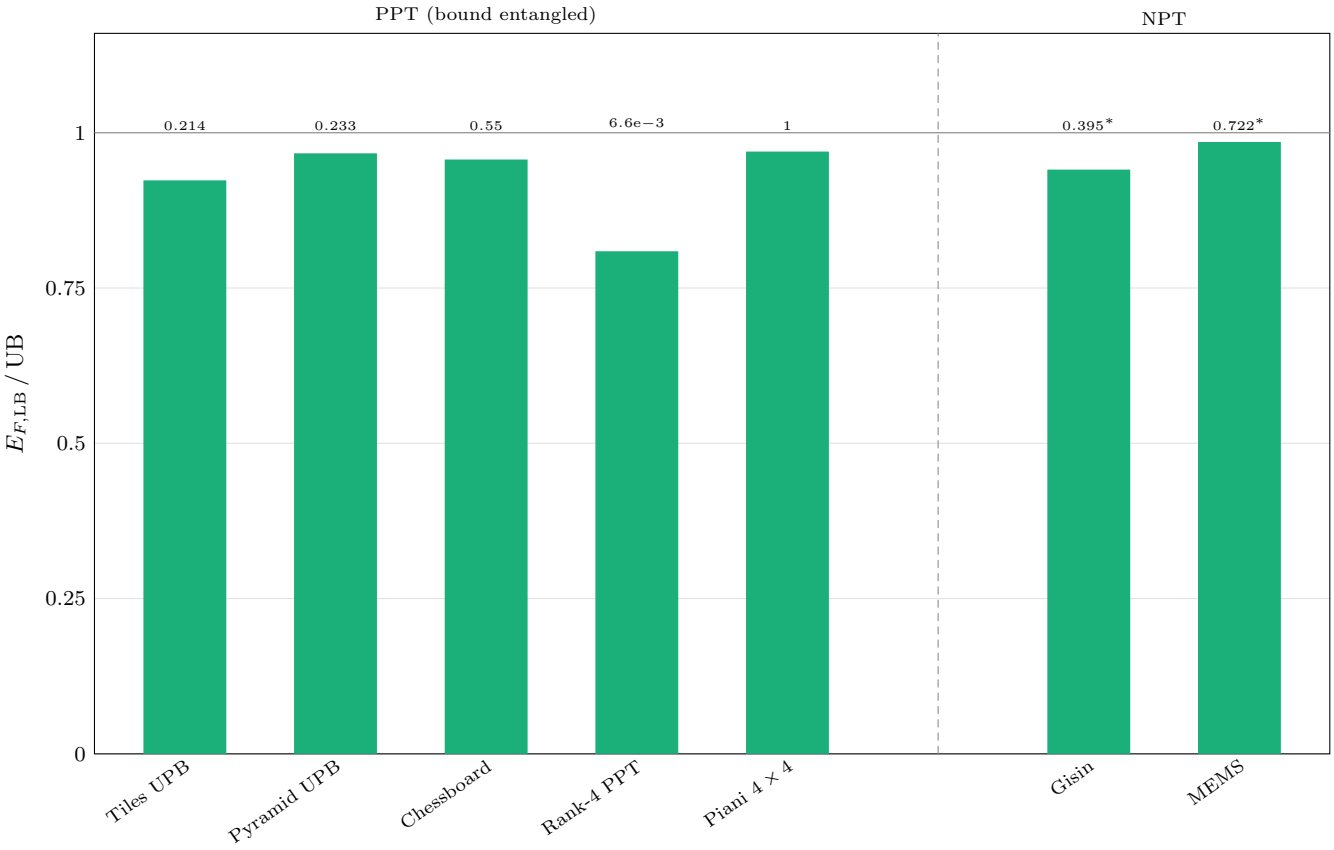


FIG. 2. Best lower bound on the entanglement of formation E_F for different PPT and NPT states, from the SDP with embedded LP at level $(k, m) = (2, 4)$, divided by the best upper bound UB printed above each state. For two qubits, UB is the exact convex roof (marked *). All levels of the hierarchy and both upper bounds, for more states, are shown in Fig. 5.

To this end, let us first recall the following criterion [50]: a pure state $|\psi\rangle$ can be converted into an ensemble $\{p_i, |\psi_i\rangle\}$ by LOCC, if and only if

$$E_j(\psi) \geq \sum_i p_i E_j(\psi_i) \qquad \text{for all } j = 1, \ldots, d_A - 1, \tag{25}$$

where E_j are the Vidal monotones (2). Hence, $|\psi\rangle$ can be converted into a mixed state ρ if and only if some decomposition of ρ satisfies all these inequalities simultaneously.

[27] More detail. Explain the Jonathan–Plenio criterion: what converting a pure state into an ensemble means (local operations with flagged outcomes), why it reduces to the Vidal monotones, and why converting into ρ only needs *some* decomposition (the flags are forgotten). A 2×2 example where the answer is known lets the reader check the method.

Notice that this is not straightforward to verify, since the decompositions of ρ maximizing each monotone E_j might be different. Therefore, to assure that all inequalities hold for the same decomposition (even if not optimal), we combine the LPs of Sec. VIII for all the E_j into a single SDP. With a grid of spectra $\{\alpha^g\}_g$, we define the following optimization:

$$s_k = \min_{\Omega_k} \max_{t} \max_{P_j \in \mathcal{P}_j} \sum_j t_j \Big(\operatorname{tr}\left((O_{P_j} \otimes \mathbb{1}^B)\Omega_k\right) - E_j(\psi) \Big), \tag{26}$$

where t is a probability vector and $\mathcal{P}_j$ is the set of polynomials of degree k satisfying the grid constraint.

[28] More detail. Give the SDP form (Eq. (G4)) in the main text as well as the min–max form, and explain in words why the weights t_j appear: the certificate is a convex combination of the violated inequalities. Mention that this SDP was infeasible for pure and low-rank targets until the polynomial coefficients were bounded ($|c| \leq 100$), and that it uses a fixed 1/60 lattice because cutting planes were too slow there.

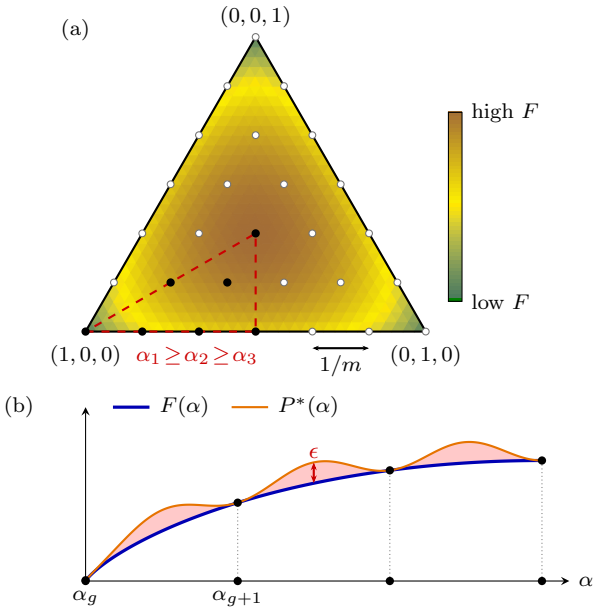


FIG. 3. Grid discretization of the simplex of spectra for $d_A = 3$. (a) Different values of $F(\alpha)$ are denoted by different colors. The dots denote the grid G, separated by a certain distance $1/m$. Since entanglement measures are symmetric functions on the spectrum, only the points with $\alpha_1 \geq \alpha_2 \geq \alpha_3$ (dashed region, filled dots) need to be imposed. (b) Cut along a one-parameter family of spectra. Although $P^* \leq F$ holds at all grid points, it is possible that $P^* > F$ between any two points by a value ϵ. This value decreases as we refine the grid.

The optimal value s_k is the smallest worst-case violation of Eq. (25) over the set of relaxations of possible decompositions. The dual variables of this SDP provide weights $t_j^* \geq 0$ with $\sum_j t_j^* = 1$ and polynomials Q_j^* satisfying $Q_j^* \leq t_j^* E_j$ on the grid (Appendix G). As shown in Sec. VII, we can extract the decomposition from the result of the SDP. This leads us to a certificate that certain pure-to-mixed LOCC transformations are possible via Eq. (25). This can be summarized as follows.

Result 6. *By extracting a decomposition $\{p_i, |\psi_i\rangle\}$, we can verify whether it satisfies Eq. (25) for a certain input state $|\psi\rangle$, and therefore determine whether they can be transformed with LOCC.*

However, unlike for the bounds for polynomial and non-polynomial measures, it may not always be possible to extract the correct decomposition. This comes from the fact that there may not exist a decomposition of ρ into linearly independent terms that satisfies the criterion of Eq. (25), which makes it very difficult for the algorithms of Sec. VII to find the underlying decomposition. Instead, a similar strategy allows us to obtain a complete no-go criterion to discard entanglement transformations.

Result 7. *We obtain a complete hierarchy for LOCC non-convertibility in the following sense:*

1. *Let $\epsilon_j = \max_{\alpha^g} \left(Q_j^*(\alpha^g) - t_j^* E_j(\alpha^g)\right)$ be the grid errors of the polynomials returned by the solver. If $s_k > \sum_j \epsilon_j$ for some k, then $|\psi\rangle$ cannot be converted into ρ by LOCC.*

2. *In the limit where the grid G is continuous and covers all possible spectra α, for any $|\psi\rangle$ and ρ, if $|\psi\rangle \overset{LOCC}{\nrightarrow} \rho$, there is a finite level k at which $s_k > 0$.*

The proof is given in Appendix G. Notice that although the grid will always be discrete in practical computations, by continuity the second item suggests that finer grids lead to tighter criteria. Numerically, we verify that this is the case for all states tested.

[29] More detail. Describe Fig. 4 in detail: the states and parameters, the region certified non-convertible, where extraction certified convertibility, and where neither test decides. Report how often the extraction succeeds, since the convertibility test is heuristic.

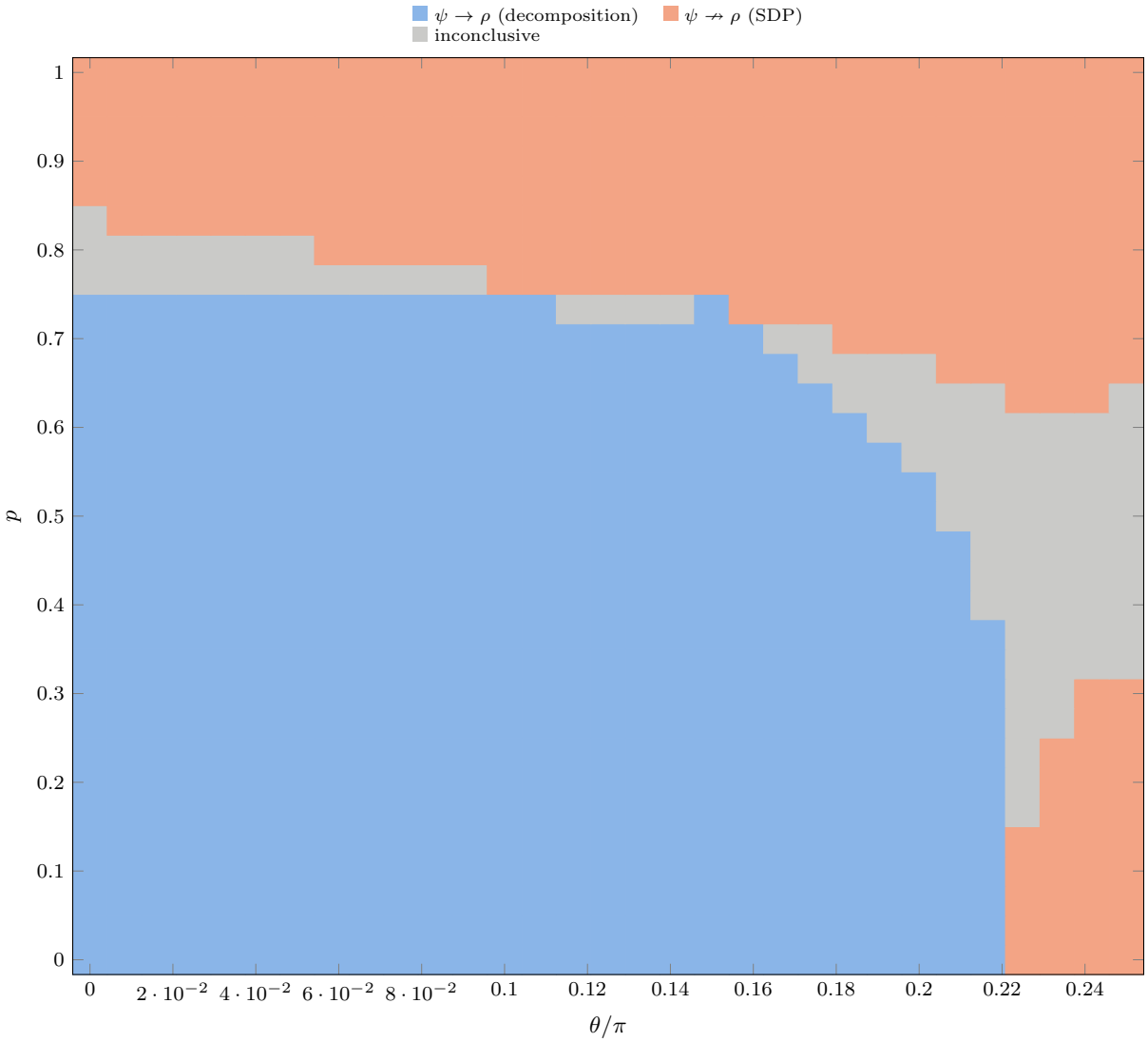


FIG. 4. LOCC convertibility from the pure state $|\psi\rangle$ with entanglement spectrum $(0.6, 0.3, 0.1)$ to the two-qutrit states $\rho(p,\theta) = p\,|\Phi_3\rangle\langle\Phi_3| + (1-p)\,|\chi_\theta\rangle\langle\chi_\theta|$, with $|\Phi_3\rangle$ maximally entangled and $|\chi_\theta\rangle = \cos\theta\,|00\rangle + \sin\theta\,|11\rangle$, from the SDP of Eq. (26) at level $(k,m) = (3,4)$.

X. CONCLUSIONS

Using spectral estimation techniques and semidefinite programming (SDP), we have introduced a method to estimate any entanglement measure for bipartite mixed states. This method also allows us to address the problem of determining whether entanglement transformations are possible from pure to mixed states, with a hierarchy of certificates of non-convertibility and a heuristic approach for convertibility. This addresses a major barrier at certifying quantum resources and their convertibility, and demonstrates the applicability of Schur-Weyl duality in this area.

For future research, it would be central to extend our approach to the multipartite case, where not all entanglement monotones are determined by the spectrum of the reduced states. The classification of convertibility between multipartite states is extremely difficult [27], but at the same time highly relevant. The reason is that relevant quantum phenomena like magic (for quantum computation) or the Fisher information (for quantum metrology) are inherently linked to multipartite systems. Extending our approach to these scenarios could be an essential step towards a full understanding of quantum resources.

[30] New content. Library chapters, before the conclusion. `schur-weyl`: what it computes (characters, isotypic projectors, Schur polynomials, Kostka numbers), the algorithms, the tests, performance (numba), version 0.4.0. `bound-entangled`: the catalogue of PPT entangled states, how each construction is checked to be PPT and entangled, the API, the tests. One page each on design choices and one short code example each are enough.

[31] More detail. Extend the conclusion: limitations (memory scaling in r and k, grid size for large d_A, heuristic extraction); the negative results (the tr_B-PPT variant, implied by the copy constraints and slower, giving 0 on every bound entangled state; random-start least squares getting stuck; extraction losing to gradient descent for E_F); open questions (multipartite case, the covariance constraint); and what you learned. Rewrite the AI statement following your programme's rules.

AI statement. The ideas, methodology, proof strategies and the project scope were designed by the authors. The numerical results, plots and proof calculations were assisted with AI (Claude Opus 4.8-5.5), and validated by the authors. The article was entirely written by the authors, with minor refinements with AI to correct spelling and condense some of the sentences.

Acknowledgements. We thank Sophia Denker, Kenneth Goodenough, Ladina Hausmann, Renato Renner and Karol Życzkowski for discussions. This research was funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation, project number 563437167), the Sino-German Center for Research Promotion (Project M-0294), and the German Federal Ministry of Research, Technology and Space (Project BeRyQC, Grant No. 13N17292).

[1] A. Einstein, B. Podolsky, and N. Rosen, "Can quantum-mechanical description of physical reality be considered complete?," *Physical Review*, vol. 47, no. 10, pp. 777–780, 1935.

[2] J. I. Cirac, A. K. Ekert, S. F. Huelga, and C. Macchiavello, "Distributed quantum computation over noisy channels," *Physical Review A*, vol. 59, no. 6, pp. 4249–4254, 1999.

[3] A. K. Ekert, "Quantum cryptography based on Bell's theorem," *Physical Review Letters*, vol. 67, no. 6, pp. 661–663, 1991.

[4] R. Raussendorf and H. J. Briegel, "A one-way quantum computer," *Physical Review Letters*, vol. 86, no. 22, pp. 5188–5191, 2001.

[5] P. W. Shor, "Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer," *SIAM Journal on Computing*, vol. 26, no. 5, pp. 1484–1509, 1997.

[6] M. A. Nielsen, "Conditions for a class of entanglement transformations," *Physical Review Letters*, vol. 83, no. 2, pp. 436–439, 1999.

[7] G. Vidal, "Entanglement monotones," *Journal of Modern Optics*, vol. 47, no. 2-3, pp. 355–376, 2000.

[8] W. H. Zurek, "Decoherence, einselection, and the quantum origins of the classical," *Reviews of Modern Physics*, vol. 75, no. 3, pp. 715–775, 2003.

[9] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems*. Oxford: Oxford University Press, 2002.

[10] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*. Cambridge: Cambridge University Press, 10th anniversary ed., 2010.

[11] C. H. Bennett, D. P. DiVincenzo, J. A. Smolin, and W. K. Wootters, "Mixed-state entanglement and quantum error correction," *Physical Review A*, vol. 54, no. 5, pp. 3824–3851, 1996.

[12] A. Uhlmann, "Entropy and optimal decompositions of states relative to a maximal commutative subalgebra," *Open Systems & Information Dynamics*, vol. 5, no. 3, pp. 209–228, 1998.

[13] A. Uhlmann, "Roofs and convexity," *Entropy*, vol. 12, no. 7, pp. 1799–1832, 2010.

[14] R. Horodecki, P. Horodecki, M. Horodecki, and K. Horodecki, "Quantum entanglement," *Reviews of Modern Physics*, vol. 81, no. 2, pp. 865–942, 2009.

[15] W. K. Wootters, "Entanglement of formation of an arbitrary state of two qubits," *Physical Review Letters*, vol. 80, no. 10, pp. 2245–2248, 1998.

[16] B. M. Terhal and K. G. H. Vollbrecht, "Entanglement of formation for isotropic states," *Physical Review Letters*, vol. 85, no. 12, pp. 2625–2628, 2000.

[17] K. G. H. Vollbrecht and R. F. Werner, "Entanglement measures under symmetry," *Physical Review A*, vol. 64, no. 6, p. 062307, 2001.

[18] F. Mintert, M. Kuś, and A. Buchleitner, "Concurrence of mixed bipartite quantum states in arbitrary dimensions," *Physical Review Letters*, vol. 92, no. 16, p. 167902, 2004.

[19] K. Chen, S. Albeverio, and S.-M. Fei, "Concurrence of arbitrary dimensional bipartite quantum states," *Physical Review Letters*, vol. 95, no. 4, p. 040504, 2005.

[20] K. Chen, S. Albeverio, and S.-M. Fei, "Entanglement of formation of bipartite quantum states," *Physical Review Letters*, vol. 95, no. 21, p. 210501, 2005.

[21] O. Gühne, M. Reimpell, and R. F. Werner, "Estimating entanglement measures in experiments," *Physical Review Letters*, vol. 98, no. 11, p. 110502, 2007.

[22] J. Eisert, F. G. S. L. Brandão, and K. M. R. Audenaert, "Quantitative entanglement witnesses," *New Journal of Physics*, vol. 9, no. 3, p. 46, 2007.

[23] G. Tóth, T. Moroder, and O. Gühne, "Evaluating convex roof entanglement measures," *Physical Review Letters*, vol. 114, no. 16, p. 160501, 2015.

[24] M. B. Plenio and S. Virmani, "An introduction to entanglement measures," *Quantum Information and Computation*, vol. 7, no. 1-2, pp. 1–51, 2007.

[25] G. Gour, "Infinite number of conditions for local mixed-state manipulations," *Physical Review A*, vol. 72, no. 2, p. 022323, 2005.

[26] E. Chitambar, D. Leung, L. Mančinska, M. Ozols, and A. Winter, "Everything you always wanted to know about LOCC (but were afraid to ask)," *Communications in Mathematical Physics*, vol. 328, no. 1, pp. 303–326, 2014.

[27] D. Sauerwein, N. R. Wallach, G. Gour, and B. Kraus, "Transformations among pure multipartite entangled states via local operations are almost never possible," *Physical Review X*, vol. 8, no. 3, p. 031020, 2018.

[28] R. Alicki, S. Rudnicki, and S. Sadowski, "Symmetry properties of product states for the system of N n-level atoms," *Journal of Mathematical Physics*, vol. 29, no. 5, pp. 1158–1162, 1988.

[29] M. Keyl and R. F. Werner, "Estimating the spectrum of a density operator," *Physical Review A*, vol. 64, no. 5, p. 052311, 2001.

[30] G. Vidal, "Entanglement of pure states for a single copy," *Physical Review Letters*, vol. 83, no. 5, pp. 1046–1049, 1999.

[31] W. Fulton and J. Harris, *Representation Theory: A First Course*, vol. 129 of *Graduate Texts in Mathematics*. New York: Springer, 1991.

[32] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*. Oxford Mathematical Monographs, Oxford: Clarendon Press, 2nd ed., 1995.

[33] A. C. Doherty, P. A. Parrilo, and F. M. Spedalieri, "Distinguishing separable and entangled states," *Physical Review Letters*, vol. 88, no. 18, p. 187904, 2002.

[34] C. M. Caves, C. A. Fuchs, and R. Schack, "Unknown quantum states: The quantum de Finetti representation," *Journal of Mathematical Physics*, vol. 43, no. 9, pp. 4537–4559, 2002.
[35] M. Christandl, R. König, G. Mitchison, and R. Renner, "One-and-a-half quantum de Finetti theorems," *Communications in Mathematical Physics*, vol. 273, no. 2, pp. 473–498, 2007.
[36] B. M. Terhal and P. Horodecki, "Schmidt number for density matrices," *Physical Review A*, vol. 61, no. 4, p. 040301, 2000.
[37] M. Christandl and G. Mitchison, "The spectra of quantum states and the Kronecker coefficients of the symmetric group," *Communications in Mathematical Physics*, vol. 261, no. 3, pp. 789–797, 2005.
[38] B. Lovitz and A. Lowe, "Nearly tight bounds for testing tree tensor network states," *IEEE Transactions on Information Theory*, vol. 72, no. 5, pp. 3074–3097, 2026.
[39] A. Tavakoli and S. Morelli, "Enhanced Schmidt-number criteria based on correlation trace norms," *Physical Review A*, vol. 110, no. 6, p. 062417, 2024.
[40] J. B. Kruskal, "Three-way arrays: rank and uniqueness of trilinear decompositions, with application to arithmetic complexity and statistics," *Linear Algebra and its Applications*, vol. 18, no. 2, pp. 95–138, 1977.
[41] R. A. Harshman, "Foundations of the PARAFAC procedure: Models and conditions for an "explanatory" multimodal factor analysis," *UCLA Working Papers in Phonetics*, vol. 16, pp. 1–84, 1970.
[42] S. E. Leurgans, R. T. Ross, and R. B. Abel, "A decomposition for three-way arrays," *SIAM Journal on Matrix Analysis and Applications*, vol. 14, no. 4, pp. 1064–1083, 1993.
[43] J. D. Carroll and J.-J. Chang, "Analysis of individual differences in multidimensional scaling via an N-way generalization of "Eckart-Young" decomposition," *Psychometrika*, vol. 35, no. 3, pp. 283–319, 1970.
[44] J. Kossaifi, Y. Panagakis, A. Anandkumar, and M. Pantic, "TensorLy: Tensor learning in Python," *Journal of Machine Learning Research*, vol. 20, no. 26, pp. 1–6, 2019.
[45] E. Schrödinger, "Probability relations between separated systems," *Mathematical Proceedings of the Cambridge Philosophical Society*, vol. 32, no. 3, pp. 446–452, 1936.
[46] E. T. Jaynes, "Information theory and statistical mechanics. II," *Physical Review*, vol. 108, no. 2, pp. 171–190, 1957.
[47] N. Gisin, "Stochastic quantum dynamics and relativity," *Helvetica Physica Acta*, vol. 62, no. 4, pp. 363–371, 1989.
[48] L. P. Hughston, R. Jozsa, and W. K. Wootters, "A complete classification of quantum ensembles having a given density matrix," *Physics Letters A*, vol. 183, no. 1, pp. 14–18, 1993.
[49] K. A. Kirkpatrick, "The Schrödinger-HJW theorem," *Foundations of Physics Letters*, vol. 19, no. 1, pp. 95–102, 2006.
[50] D. Jonathan and M. B. Plenio, "Minimal conditions for local pure-state entanglement manipulation," *Physical Review Letters*, vol. 83, no. 7, pp. 1455–1458, 1999.
[51] O. Bucicovschi and J. Lebl, "On the continuity and regularity of convex extensions," *Journal of Convex Analysis*, vol. 20, no. 4, pp. 1113–1126, 2013.
[52] M. Navascués, M. Owari, and M. B. Plenio, "Power of symmetric extensions for entanglement detection," *Physical Review A*, vol. 80, no. 5, p. 052306, 2009.
[53] M. Sion, "On general minimax theorems," *Pacific Journal of Mathematics*, vol. 8, no. 1, pp. 171–176, 1958.
[54] K. He, J. Saunderson, and H. Fawzi, "QICS: Quantum information conic solver," 2024. arXiv:2410.17803.
[55] A. Van Cleemput, "bound-entangled: Reference implementations of bound entangled states." `https://github.com/AVlyx/bound-entangled`, 2026. Python package, version 0.2.2.
[56] V. Russo, "toqito – Theory of quantum information toolkit: A Python package for studying quantum information," *Journal of Open Source Software*, vol. 6, no. 61, p. 3082, 2021.
[57] B. Röthlisberger, J. Lehmann, and D. Loss, "Numerical evaluation of convex-roof entanglement measures with applications to spin rings," *Physical Review A*, vol. 80, no. 4, p. 042301, 2009.

Appendix A: Proof of Result 1

We consider a polynomial P of degree m. At level $k \geq m$ of the hierarchy, we obtain the optimal solution Ω_k, with $P_{\mathrm{LB}_k}(\rho) = \mathrm{tr}(O_P\,\Omega_m)$ where $\Omega_m = \mathrm{tr}_{m+1,\ldots,k}(\Omega_k)$ is the m-copy marginal of Ω_k. By the de Finetti theorem for Bose-symmetric states [35, Thm. II.8], for every feasible Ω_k, there is an $\omega_m = \int |\phi\rangle\langle\phi|^{\otimes m}\, d\mu(\phi)$ such that

$$\|\Omega_m - \omega_m\|_1 \leq \epsilon \tag{A1}$$

with $\epsilon = \frac{4Dm}{k}$. By Carathéodory's theorem, since the space is finite-dimensional, we can write $\omega_m = \sum_i p_i\, |\phi_i\rangle\langle\phi_i|^{\otimes m}$. Here, ω_m is associated to a decomposition $\{p_i, |\phi_i\rangle\}_i$ corresponding to a state σ and $\mathrm{tr}(O_P\,\omega_m) \geq \widehat{P}(\sigma)$. Since the partial trace is contractive, $\|\rho - \sigma\|_1 = \|\mathrm{tr}_{2,\ldots,m}(\Omega_m - \omega_m)\|_1 \leq \epsilon$.

Then, by Hölder inequality (A2), by definition of the convex roof (A3) and from the continuity of the convex roof of a continuous function in finite dimensions (A4) [51, Thm. 6.1],

$$\begin{aligned}
\mathrm{tr}(O_P\Omega_m) &\geq \mathrm{tr}(O_P\omega_m) - \epsilon\|O_P\|_\infty && \text{(A2)}\\
&\geq \widehat{P}(\sigma) - \epsilon\|O_P\|_\infty && \text{(A3)}\\
&\geq \widehat{P}(\rho) - \delta(\epsilon) - \epsilon\|O_P\|_\infty && \text{(A4)}
\end{aligned}$$

with $\delta(\epsilon) \to 0$.

Hence, $\widehat{P}(\rho) \geq P_{\mathrm{LB}_k}(\rho) \geq \widehat{P}(\rho) - \delta(\epsilon) - \epsilon\|O_P\|_\infty$, and therefore $P_{\mathrm{LB}_k}(\rho) \to \widehat{P}(\rho)$ as $k \to \infty$.

[32] Deeper proof. This is the main theorem; expand the proof. State the de Finetti theorem precisely (hypotheses, norm, constant) and check $\epsilon = 4Dm/k$ against Ref. [35], with $D = d_A d_B$ as the single-copy dimension and Ω_k normalised by the marginal constraint. Explain why Carathéodory turns the integral into a finite sum. State the continuity theorem of Ref. [51] that you use. Bound $\|O_P\|_\infty \leq \max_\lambda |c_\lambda|/f^\lambda$ (the Π^λ are orthogonal). If possible, give an explicit rate in k for one measure, e.g. the linear entropy, using an explicit continuity bound for its convex roof.

Appendix B: Lifting

Lemma 8. *Let P be a homogeneous symmetric polynomial of degree m with $m < k$. Let P_k be its lift to degree k such that $P_k(\alpha) = P(\alpha)\left(\sum_i \alpha_i\right)^{k-m} = P(\alpha)$ where α is the spectrum of a state ρ_A. Then, for every Ω_k acting on $(\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B})^{\otimes k}$, which is permutation invariant across all k copies of $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$,*

$$\mathrm{tr}\Big(O_P^{A_1,\ldots,A_m} \otimes \mathbb{1}^{A_{m+1},\ldots,A_k} \otimes \mathbb{1}^B \Omega_k\Big) = \mathrm{tr}\big(O_{P_k}^A \otimes \mathbb{1}^B \Omega_k\big). \tag{B1}$$

Proof. Let $\mathcal{T} = \frac{1}{k!}\sum_{\pi \in S_k} \pi\ O_P \otimes \mathbb{1}^{\otimes(k-m)}\ \pi^\dagger$ be the twirl of $O_P \otimes \mathbb{1}^{\otimes(k-m)}$ over S_k. Obviously $\mathcal{T}$ commutes with every permutation in S_k, and, since it is a linear combination of permutations, we can write it as a sum of isotypic projectors $\mathcal{T} = \sum_{\lambda \vdash k} a_\lambda \Pi^\lambda$ for some coefficients a_λ.

For any ρ_A with spectrum α, since $\rho_A^{\otimes k}$ is permutation invariant,

$$\mathrm{tr}\big(\mathcal{T}\rho_A^{\otimes k}\big) = \mathrm{tr}\left(\frac{1}{k!}\sum_{\pi \in S_k} \pi O_P \otimes \mathbb{1}\pi^\dagger \rho_A^{\otimes k}\right) = \mathrm{tr}\big(O_P \otimes \mathbb{1}\rho_A^{\otimes k}\big). \tag{B2}$$

Thus $\mathcal{T}$ and O_{P_k} have the same expectation value $P_k(\alpha)$ on every $\rho_A^{\otimes k}$, and since both are combinations of isotypic projectors and the Schur polynomials are linearly independent, $\mathcal{T} = O_{P_k}$. Finally,

$$\mathrm{tr}\Big(O_P^{A_1,\ldots,A_m} \otimes \mathbb{1}^{A_{m+1},\ldots,A_k} \otimes \mathbb{1}^B \Omega_k\Big) \tag{B3}$$

$$= \mathrm{tr}\left(\frac{1}{k!}\sum_{\pi \in S_k} O_P^{A_1,\ldots,A_m} \otimes \mathbb{1}^{A_{m+1},\ldots,A_k} \otimes \mathbb{1}^B \pi^\dagger \Omega_k \pi\right) \tag{B4}$$

$$= \mathrm{tr}\left(\frac{1}{k!}\sum_{\pi \in S_k} \pi O_P^{A_1,\ldots,A_m} \otimes \mathbb{1}^{A_{m+1},\ldots,A_k} \otimes \mathbb{1}^B \pi^\dagger \Omega_k\right) \tag{B5}$$

$$= \mathrm{tr}\big(O_{P_k} \otimes \mathbb{1}^B \Omega_k\big) \tag{B6}$$

□

[33] Deeper proof. Two steps deserve a sentence each: why $O_P \otimes \mathbb{1}^{\otimes(k-m)}$ is a linear combination of permutations of S_k (embed $S_m \subset S_k$), and why a permutation-invariant element of the span of permutations is a combination of the Π^λ (the centre of the group algebra is spanned by the class sums, equivalently by the central idempotents). Also note that both s_λ and Π_A^λ vanish for $\ell(\lambda) > d_A$, so linear independence is only needed for $\ell(\lambda) \leq d_A$.

Appendix C: Monotonicity in the level

In the full SDP, at level $k+1$, we obtain the optimal solution Ω_{k+1}. Let O_k be the observable at level k and O_{k+1} be its lift to level $k+1$. Then, $P_{\mathrm{LB}_{k+1}} = \mathrm{tr}(O_{k+1}\Omega_{k+1}) = \mathrm{tr}(O_k \otimes \mathbb{1}\ \Omega_{k+1})$ according to Lemma 8. At level k, $\mathrm{tr}_{A_{k+1}B_{k+1}}(\Omega_{k+1})$ is also in the feasible set – PPT across copies, permutation invariant, first marginal equal to ρ and PSD – with $\mathrm{tr}\big(O_k\,\mathrm{tr}_{A_{k+1}B_{k+1}}(\Omega_{k+1})\big) = P_{\mathrm{LB}_{k+1}}$. Then, $P_{\mathrm{LB}_{k+1}} \geq P_{\mathrm{LB}_k}$. We can use the same argument in k and m for the simplified SDP.

[34] Deeper proof. Write out the simplified case instead of "the same argument": which registers are traced out to go from $(k, m+1)$ to (k, m) and from $(k+1, m)$ to (k, m), and which constraints survive each step.

Appendix D: Symmetry reductions and optimizations

Throughout this appendix, a single copy of $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ is identified with $\mathbb{C}^D$, with orthonormal basis $\{|j\rangle\}_{j=0}^{D-1}$.

1. Restriction to the symmetric subspace

The symmetric subspace $\mathrm{Sym}^k(\mathbb{C}^D)$ has an orthonormal basis indexed by occupation vectors. The occupation vector denoted $\nu(i_1, \ldots, i_k) \in \mathbb{N}^D$ of a basis vector $|i_1, \ldots, i_k\rangle \in (\mathbb{C}^D)^{\otimes k}$ has entries ν_j equal to the number of times j appears in $(i_1, \ldots, i_k)$. For example, $\nu(0, 1, 2, 0, 0, 1, 2, 2) = (3, 2, 3)$ for $D = 3$, $k = 8$. We define the isometry $V_k : (\mathbb{C}^D)^{\otimes k} \to \mathrm{Sym}^k(\mathbb{C}^D)$ as

$$V_k = \sum_{i_1,\ldots,i_k=0}^{D-1} \binom{k}{\nu(i_1, \ldots, i_k)}^{-1/2} |\nu(i_1, \ldots, i_k)\rangle \langle i_1, \ldots, i_k| , \tag{D1}$$

where $\binom{k}{\nu} = k!/\prod_j \nu_j!$ is the multinomial coefficient, i.e., the number of basis vectors with occupation vector ν. It has real entries and satisfies $V_k V_k^\dagger = \mathbb{1}_{\mathrm{Sym}^k}$ and $V_k^\dagger V_k = \Pi_{\mathrm{sym}}^k$. Since every feasible Ω_k is supported on the symmetric subspace, $\Omega_k = V_k^\dagger \Omega_k^{\mathrm{sym}} V_k$ with $\Omega_k^{\mathrm{sym}} = V_k \Omega_k V_k^\dagger$, a matrix of size $\binom{k+D-1}{k}$. The objective becomes $\mathrm{tr}\left(V_k (O_P \otimes \mathbb{1}^B) V_k^\dagger \, \Omega_k^{\mathrm{sym}}\right)$.

2. First marginal constraint

Here we wish to encode the condition $\mathrm{tr}_{A_2 B_2 \ldots A_k B_k}(\Omega_k) = \rho$. We define the occupation vector $\nu_\downarrow$ such that $|\nu_\downarrow\rangle \in \mathrm{Sym}^{k-1}(\mathbb{C}^D)$, and $|\nu_\downarrow + e_j\rangle \in \mathrm{Sym}^k(\mathbb{C}^D)$ has one more element $|j\rangle$, i.e., $(\nu_\downarrow + e_j)_j = (\nu_\downarrow)_j + 1$. Grouping the $\binom{k-1}{\nu_\downarrow}$ tuples $(i_2, \ldots, i_k)$ with the same occupation vector and using $\binom{k}{\nu_\downarrow + e_j} = \binom{k-1}{\nu_\downarrow} \frac{k}{\nu_{\downarrow,j}+1}$, we get

$$(\mathrm{tr}_{2,\ldots,k} \Omega_k)_{jj'} = \sum_{\nu_\downarrow} \binom{k-1}{\nu_\downarrow} \frac{\langle \nu_\downarrow + e_j | \, \Omega_k^{\mathrm{sym}} \, | \nu_\downarrow + e_{j'} \rangle}{\sqrt{\binom{k}{\nu_\downarrow + e_j}\binom{k}{\nu_\downarrow + e_{j'}}}} = \frac{1}{k} \mathrm{tr}\left(\hat{a}_{j'}^\dagger \hat{a}_j \, \Omega_k^{\mathrm{sym}}\right), \tag{D2}$$

with the bosonic creation operators $\hat{a}_j^\dagger : \mathrm{Sym}^{k-1}(\mathbb{C}^D) \to \mathrm{Sym}^k(\mathbb{C}^D)$,

$$\hat{a}_j^\dagger = \sum_{\nu_\downarrow} \sqrt{\nu_{\downarrow,j} + 1} \; |\nu_\downarrow + e_j\rangle \langle \nu_\downarrow| . \tag{D3}$$

Thus, the marginal constraint becomes the linear constraint $\frac{1}{k} \sum_{j,j'} \mathrm{tr}\left(\hat{a}_{j'}^\dagger \hat{a}_j \, \Omega_k^{\mathrm{sym}}\right) |j\rangle \langle j'| = \rho$.

3. Partial-transpose constraints

Thanks to the permutation symmetry, the partial transpose only depends on the size of the set of transposed copies, and since $\Omega_k^{T_{S^c}} = (\Omega_k^{T_S})^T$, a set and its complement give equivalent constraints. We therefore impose the PPT constraints on the first l copies for $l \in \{1, \ldots, \lfloor \frac{k}{2} \rfloor\}$. We define the isometry

$$W_l = (V_l \otimes V_{k-l}) V_k^\dagger : \; \mathrm{Sym}^k(\mathbb{C}^D) \to \mathrm{Sym}^l(\mathbb{C}^D) \otimes \mathrm{Sym}^{k-l}(\mathbb{C}^D), \tag{D4}$$

so that $\Omega_k = (V_l \otimes V_{k-l})^\dagger \, W_l \Omega_k^{\mathrm{sym}} W_l^\dagger \, (V_l \otimes V_{k-l})$. Writing T_l also for the transpose of the factor $\mathrm{Sym}^l(\mathbb{C}^D)$ in the occupation basis, the partial transpose becomes

$$\Omega_k^{T_l} = \left((V_l \otimes V_{k-l})^\dagger \, W_l \Omega_k^{\mathrm{sym}} W_l^\dagger \, (V_l \otimes V_{k-l})\right)^{T_l} \tag{D5}$$

$$= \left(V_l^T \otimes V_{k-l}^\dagger\right) \left(W_l \Omega_k^{\mathrm{sym}} W_l^\dagger\right)^{T_l} (V_l^* \otimes V_{k-l}) \tag{D6}$$

$$= (V_l \otimes V_{k-l})^\dagger \left(W_l \Omega_k^{\mathrm{sym}} W_l^\dagger\right)^{T_l} (V_l \otimes V_{k-l}), \tag{D7}$$

since V_l is real, so that $V_l^T = V_l^\dagger$ and $V_l^* = V_l$. Conjugating back with $V_l \otimes V_{k-l}$, the condition becomes

$$\Omega_k^{T_l} \succeq 0 \iff \left(W_l\, \Omega_k^{\mathrm{sym}} W_l^\dagger\right)^{T_l} \succeq 0, \qquad 1 \le l \le \lfloor k/2 \rfloor. \tag{D8}$$

This is the most expensive constraint of the SDP, since it encodes the state in the larger space $\mathrm{Sym}^l(\mathbb{C}^D) \otimes \mathrm{Sym}^{k-l}(\mathbb{C}^D)$ to support the partial transpose. While these constraints are not necessary for the hierarchy to be complete, they help convergence at finite k, as shown in Ref. [52].

[35] More detail. Add a small worked example of the occupation basis ($D = 2$, $k = 2$: the three states $|2,0\rangle, |1,1\rangle, |0,2\rangle$ and V_2 as a 3×4 matrix). Quantify what the PPT constraints cost and gain: bounds with and without them at the same level, and the size of the $\mathrm{Sym}^l \otimes \mathrm{Sym}^{k-l}$ blocks.

4. Real variables

If ρ is real, the SDP can be restricted to real symmetric matrices. If Ω_k is feasible, so is its entrywise complex conjugate Ω_k^*, since conjugation commutes with partial traces, partial transposes and the real projector Π_{sym}^k, and $\rho^* = \rho$. Since the isotypic projectors are linear combinations of permutation matrices with real characters, O_P is real, and

$$\mathrm{tr}\left((O_P^A \otimes \mathbb{1}^B)\,\Omega_k\right) = \mathrm{tr}\left((O_P^A \otimes \mathbb{1}^B)\,\Omega_k^*\right) = \mathrm{tr}\left((O_P^A \otimes \mathbb{1}^B)\,\tfrac{\Omega_k + \Omega_k^*}{2}\right). \tag{D9}$$

By convexity, $\frac{1}{2}(\Omega_k + \Omega_k^*)$ is feasible, real and has the same objective value, so the optimum is attained by a real matrix. For the true ω_k, this real matrix is $\frac{1}{2}\sum_i p_i \left(|\psi_i\rangle\langle\psi_i|^{\otimes k} + |\psi_i^*\rangle\langle\psi_i^*|^{\otimes k}\right)$, which corresponds to the decomposition $\{\frac{p_i}{2}, |\psi_i\rangle\} \cup \{\frac{p_i}{2}, |\psi_i^*\rangle\}$ of ρ. The same argument applies to $\Phi_{k,m}$ in the simplified relaxation.

[36] More detail. Add the remark that a real ρ does not imply a real optimal decomposition: the argument only shows that the optimal Ω_k can be taken real, which corresponds to the symmetrised ensemble with $|\psi_i\rangle$ and $|\psi_i^*\rangle$. Quantify the saving in memory and time on one example.

5. Projection onto the support of ρ

Let $r = \mathrm{rank}\,\rho$, and let $V_\rho : \mathbb{C}^D \to \mathbb{C}^r$ be the isometry onto the support of ρ, given by the eigenvectors of ρ with nonzero eigenvalue, so that $V_\rho V_\rho^\dagger = \mathbb{1}_r$ and $V_\rho^\dagger V_\rho = \Pi_\rho$, the projector onto the support of ρ. Every feasible Ω_k is supported on $(\mathrm{supp}\,\rho)^{\otimes k}$: since

$$\mathrm{tr}\left(((\mathbb{1} - \Pi_\rho) \otimes \mathbb{1})\,\Omega_k\right) = \mathrm{tr}\left((\mathbb{1} - \Pi_\rho)\rho\right) = 0 \tag{D10}$$

and $\Omega_k \succeq 0$, we have $((\mathbb{1} - \Pi_\rho) \otimes \mathbb{1})\,\Omega_k = 0$, and by permutation symmetry the same holds on every copy. Hence $\Omega_k = \Pi_\rho^{\otimes k} \Omega_k \Pi_\rho^{\otimes k}$, and

$$\mathrm{tr}\left((O_P^A \otimes \mathbb{1}^B)\,\Omega_k\right) = \mathrm{tr}\left(\left(V_\rho^{\otimes k}(O_P^A \otimes \mathbb{1}^B)V_\rho^{\dagger\otimes k}\right)\left(V_\rho^{\otimes k}\Omega_k V_\rho^{\dagger\otimes k}\right)\right). \tag{D11}$$

We can therefore equivalently optimize over $\Omega_k^\rho = V_\rho^{\otimes k}\Omega_k V_\rho^{\dagger\otimes k}$ on $(\mathbb{C}^r)^{\otimes k}$ with marginal $V_\rho \rho V_\rho^\dagger$. Bose symmetry is preserved, since $V_\rho^{\otimes k}$ commutes with permutations, and the partial-transpose constraints are equivalent, since $(\Omega_k^\rho)^{T_l}$ and $\Omega_k^{T_l}$ are related in both directions by congruences with V_ρ and V_ρ^*. Combined with the symmetric subspace, the variable has size $\binom{k+r-1}{k}$. If ρ is real, V_ρ can be chosen real and the real-variable reduction still applies.

The same argument applies to the simplified relaxation: by the permutation invariance of $\mathrm{tr}_B\,\Phi_{k,m}$, every A register has reduced state $\rho_A = \mathrm{tr}_B(\rho)$, and therefore all A registers can be encoded on the support of ρ_A.

Appendix E: Tensor decomposition

1. Heuristic extraction

We take the optimal solution Ω_k of Eq. (17). We work on the support of ρ as in Appendix D 5: with $r = \mathrm{rank}\,\rho$ and the isometry $V_\rho : \mathrm{supp}\,\rho \to \mathbb{C}^r$, we consider $\Omega_\rho = V_\rho^{\otimes k}\Omega_k V_\rho^{\dagger\otimes k}$.

To work with real tensors, let $\{G_a\}_{a=0}^{r^2-1}$ be a Hilbert–Schmidt orthonormal basis of Hermitian operators on $\mathbb{C}^r$, consisting of the normalized identity $G_0 = \mathbb{1}/\sqrt{r}$ and the generalized Gell-Mann matrices. Every pure state can be written as $|\psi\rangle\langle\psi| = \sum_a x_a G_a$ with a real vector x satisfying $x_0 = 1/\sqrt{r}$ and $\|x\|_2 = 1$. We define the real symmetric order-k tensor $\Omega^G_{a_1\dots a_k} = \operatorname{tr}\left((G_{a_1}\otimes\cdots\otimes G_{a_k})\,\Omega_\rho\right)$, which for the true ω_k equals $\sum_i p_i\, x^{(i)\otimes k}$. We run PARAFAC (alternating least squares) on Ω^G with nonnegative weights, obtaining terms $\{p_i, v_i\}$; we normalize each factor such that $v_{i,0} = 1/\sqrt{r}$, absorbing the scale into p_i, and recover the pure states $|\psi_i\rangle$ as the top eigenvectors of $\sum_a v_{i,a} G_a$. Finally, we repair the decomposition $\{p_i, |\psi_i\rangle\}$ as described in Sec. VII.

The number of terms n fitted by PARAFAC is the rank of Ω_ρ, which equals the number of terms when the $|\psi_i\rangle\langle\psi_i|^{\otimes k}$ are linearly independent. Any decomposition of ρ has at least r terms, otherwise the repair fails. Conversely, every decomposition of ρ consists of states on $\operatorname{supp}\rho \cong \mathbb{C}^r$, whose state space has dimension r^2-1, and therefore by Carathéodory's theorem the convex roof of a continuous function is attained by a decomposition with at most r^2-1 terms [12, Lemma A-2].

[37] Deeper proof. Check the count: Carathéodory in a convex set of dimension n gives $n+1$ points, so the (r^2-1)-dimensional state space gives r^2 terms, which is also what Uhlmann's lemma and App. I 1 ($M = r^2$) use. The text says r^2-1; make the two consistent and write out the argument.

Before the extraction, it helps to reduce the rank of the SDP solution. We solve a second SDP, minimizing $\operatorname{tr}(R_3 \otimes \mathbb{1}\,\Omega_k)$ for a random Hermitian R_3 acting on the first 3 copies, over the same feasible set with the additional constraint $\operatorname{tr}\left((O_P \otimes \mathbb{1}^B)\Omega_k\right) \le P_{\mathrm{LB}} + \delta_k$ for a small $\delta_k \to 0$ with $\delta_k \ge \widehat{P}(\rho) - P_{\mathrm{LB}_k}$. This selects an extreme point of the near-optimal face, which generically has low rank, instead of a mixture of several optimal solutions.

[38] Deeper proof. The condition $\delta_k \ge \widehat{P}(\rho) - P_{\mathrm{LB}_k}$ involves the unknown convex roof. Say how δ_k is chosen in practice (e.g. from the gap to an upper bound, which guarantees it), mention that the objective had to be normalised for QICS to converge, and justify "an extreme point generically has low rank" (cite or sketch the rank bound for extreme points of spectrahedra).

2. Convergence of the extraction

Given a state $\omega_3 = \sum_i p_i\, |\psi_i\rangle\langle\psi_i|^{\otimes 3}$, we can use Jennrich's algorithm to recover uniquely the decomposition $\{p_i, |\psi_i\rangle\}_i$ when the $|\psi_i\rangle\langle\psi_i|$ are linearly independent. Therefore, we need to select a low-rank extreme point $\Omega_3 = \operatorname{tr}_{A_4B_4\dots A_kB_k}(\Omega_k)$. For this we apply the rank reduction step as shown in App. E 1.

Now we need the following observation.

Observation 9. *A decomposition $\{q_i, |\psi_i\rangle\langle\psi_i|\}$ of ρ is extremal in the set of optimal decompositions of ρ with regards to a polynomial P, only if $|\psi_i\rangle\langle\psi_i|$ are linearly independent projectors.*

Proof. We will do so by contradiction as follows. Given an optimal decomposition $\{q_i, |\psi_i\rangle\langle\psi_i|\}$ of ρ, if the projectors $|\psi_i\rangle\langle\psi_i|$ are linearly dependent, there exists a nonzero real vector c such that $\sum_i c_i\, |\psi_i\rangle\langle\psi_i| = 0$ with $\sum_i c_i = 0$. Assuming that this is the case, for a small $0 < t \le \min_i \frac{q_i}{|c_i|}$, let us define two other decompositions $\{q_i + tc_i, |\psi_i\rangle\langle\psi_i|\}$ and $\{q_i - tc_i, |\psi_i\rangle\langle\psi_i|\}$ such that their midpoint is the decomposition $\{q_i, |\psi_i\rangle\langle\psi_i|\}$. Since $\sum_i (q_i + tc_i)P(\psi_i) \ge \widehat{P}(\rho) = \sum_i q_i P(\psi_i)$ and $\sum_i (q_i - tc_i)P(\psi_i) \ge \widehat{P}(\rho)$, then $\sum_i (q_i + tc_i)P(\psi_i) = \sum_i (q_i - tc_i)P(\psi_i) = \widehat{P}(\rho)$. These decompositions are also optimal and their midpoint is the decomposition extracted $\{q_i, |\psi_i\rangle\langle\psi_i|\}$, which proves that it is not an extreme point. Therefore, any extreme point of the convex set of optimal decompositions must have linearly independent projectors. □

For almost every R, the minimizer ω_3^* of $\operatorname{tr}(R\,\omega_3)$ among the ω_3 of optimal decompositions is unique. The decomposition of ρ giving ω_3^* has linearly independent projectors. Using the same argument shown above, if it has linearly dependent terms, we can define two decompositions whose associated ω_3 is optimal with midpoint ω_3^*. By uniqueness, both decompositions give ω_3^*. We can simply increase t until a weight vanishes and repeat to end up at a decomposition into linearly independent terms. Therefore, ω_3^* decomposes into linearly independent terms.

As $k \to \infty$ the rank-reduced Ω_3 converges to $\omega_3(\mu_e) = \sum_i p_i\, |\psi_i\rangle\langle\psi_i|^{\otimes 3}$, where μ_e is the extreme optimal decomposition singled out by R, since $\delta_k \ge \widehat{P}(\rho) - P_{\mathrm{LB}_k}$ leaves $\omega_3(\mu_e)$ feasible during the rank reduction step. By App. A, every limit point as $k \to \infty$ corresponds to the state ω_3 of an optimal decomposition, and for almost every R the minimizer of $\operatorname{tr}(R\,\cdot)$ on this set is unique and has linearly independent projectors. Then, Jennrich's algorithm is continuous at $\omega_3(\mu_e)$, and therefore applied to Ω_3 it gives a decomposition of a state σ that is ϵ-close to ρ with $\epsilon \to 0$ as $k \to \infty$, which after the repair step has value converging to $\widehat{P}(\rho)$.

This argument does not extend to LOCC since the optimal decompositions do not form a face of the set of decompositions of ρ, and the extreme points may not have linearly independent projectors, in which case we cannot use Jennrich's algorithm.

[39] Deeper proof. This is the least detailed proof in the paper and worth the most care. (i) In the observation, say why $\sum_i c_i = 0$: take the trace of $\sum_i c_i |\psi_i\rangle\langle\psi_i| = 0$. (ii) Justify that the minimiser of $\mathrm{tr}(R\,\cdot)$ is unique for almost every R (a compact convex set has a unique minimiser outside a measure-zero set of directions). (iii) Justify that Jennrich's algorithm is continuous at $\omega_3(\mu_e)$: generic distinct eigenvalues of the random contraction plus eigenvector perturbation bounds. (iv) Jennrich needs the terms to be linearly independent in a space of dimension r^2, consistent with the Carathéodory count above. (v) Write "increase t until a weight vanishes and repeat" as an induction on the number of terms.

Appendix F: Linear program for non-polynomial functions

1. Construction of the LP and of the single SDP

We optimize over symmetric polynomials of degree at most k. Since polynomials of lower degree can be lifted to degree k (Lemma 8), we parametrize them as $P(\alpha) = \sum_{\lambda\vdash k} c_\lambda s_\lambda(\alpha)$, and the LP variables are the coefficients $\vec{c} = (c_\lambda)_\lambda$. Given the function F, we require $P(\alpha) \le F(\alpha)$ for all spectra α, and enforce this constraint discretely on a grid of spectra $\{\alpha^g\}_g$, where the values $s_\lambda(\alpha^g)$ and $F(\alpha^g)$ are precomputed. By symmetry, it suffices to take grid points with $\alpha_{g,1} \ge \alpha_{g,2} \ge \cdots \ge \alpha_{g,d_A} \ge 0$ and $\sum_i \alpha_{g,i} = 1$.

Writing $\mu_\lambda(\Phi) = \mathrm{tr}\left((\frac{1}{f^\lambda}\Pi^\lambda \otimes \mathbb{1}^B)\Phi\right)$, the objective is $\mathrm{tr}\left((O_P \otimes \mathbb{1}^B)\Omega_k\right) = \sum_\lambda c_\lambda\, \mu_\lambda(\Omega_k)$, and therefore the bound reads

$$F_{\mathrm{LB}_k} = \max_{\vec{c}} \min_{\Omega_k} \sum_\lambda c_\lambda\, \mu_\lambda(\Omega_k) \quad \text{s.t.} \quad \sum_\lambda c_\lambda s_\lambda(\alpha^g) \le F(\alpha^g) \ \ \forall g, \tag{F1}$$

with Ω_k ranging over the feasible set of Eq. (17). Since the objective is bilinear, the set of feasible Ω_k is convex and compact, and the set of valid $\vec{c}$ is convex, we can exchange the maximum and the minimum [53]. Then, for fixed Ω_k, the inner maximization is an LP whose dual is

$$\min_{y\ge 0} \sum_g y_g F(\alpha^g) \quad \text{s.t.} \quad \sum_g y_g\, s_\lambda(\alpha^g) = \mu_\lambda(\Omega_k) \ \ \forall\lambda. \tag{F2}$$

The primal is always feasible, since the constant polynomial $P = \min_g F(\alpha^g)$ is admissible, so strong duality holds. We merge the LP and the SDP to obtain a single SDP,

$$\begin{aligned} F_{\mathrm{LB}_k} = \min_{\Omega_k,\, y} & \sum_g y_g F(\alpha^g) \\ \text{s.t. } & \Omega_k \text{ feasible for Eq. (17)}, \quad y \ge 0, \\ & \sum_g y_g\, s_\lambda(\alpha^g) = \mathrm{tr}\left((\tfrac{1}{f^\lambda}\Pi^\lambda \otimes \mathbb{1}^B)\,\Omega_k\right) \quad \forall\lambda \vdash k. \end{aligned} \tag{F3}$$

The coefficients c_λ of the optimal polynomial are recovered as the dual variables of the equality constraints. Since the grid is discrete, the optimal polynomial P^* may exceed F between grid points. We therefore report $F_{\mathrm{LB}_k} - \epsilon$, where $\epsilon = \max_\alpha (P^*(\alpha) - F(\alpha))$ is estimated from the polynomial returned by the solver. This is a valid lower bound, since $P^* - \epsilon \le F$ on all spectra.

2. Convergence of the bound

Here we prove Result 5 for the bound $\widetilde{F}_{\mathrm{LB}_k}$ of Eq. (22), in which $P \le F$ is imposed on all spectra. Monotonicity follows naturally from the facts that admissible polynomials stay admissible at higher level by lifting (Lemma 8), and that the SDP is monotone (Appendix C).

By the Weierstrass approximation theorem, for any continuous function F' and any $\eta > 0$, we can find a polynomial P' such that $|F'(\alpha) - P'(\alpha)| < \eta$ for all spectra α. Then, by averaging over permutations of the variables, for any continuous symmetric function F, we can find a symmetric polynomial P such that $|F(\alpha) - P(\alpha)| < \eta$ for all spectra α. Shifting P by η, for any F and any $\eta > 0$, there exists a symmetric polynomial P such that $P(\alpha) < F(\alpha)$ and

$F(\alpha) - P(\alpha) < 2\eta$ for all spectra α. This P is admissible for $\deg P \le k$, and therefore by Result 1, $\lim_{k\to\infty} \widetilde{F}_{\mathrm{LB}_k}(\rho) \ge \lim_{k\to\infty} P_{\mathrm{LB}_k}(\rho) = \widehat{P}(\rho) \ge \widehat{F}(\rho) - 2\eta$. Conversely, $P \le F$ implies $\widehat{P}(\rho) \le \widehat{F}(\rho)$ for every admissible P, and therefore $\widetilde{F}_{\mathrm{LB}_k}(\rho) \le \widehat{F}(\rho)$. Since η is arbitrary,

$$\lim_{k\to\infty} \widetilde{F}_{\mathrm{LB}_k}(\rho) = \widehat{F}(\rho). \tag{F4}$$

With a finite grid, the set of admissible polynomials is larger, and therefore $F_{\mathrm{LB}_k} \ge \widetilde{F}_{\mathrm{LB}_k}$, and $F_{\mathrm{LB}_k} - \epsilon \le \widehat{F}(\rho)$, where ϵ is the grid error of the returned polynomial. The uncorrected value F_{LB_k} is nondecreasing in the level by the same argument as above, but the corrected value $F_{\mathrm{LB}_k} - \epsilon$ need not be, since the grid error can change between levels. In our numerical results, we observe that increasing the level yields tighter bounds.

[40] Deeper proof. Sion's theorem needs one of the two sets to be compact; the set of feasible Ω_k is, but say so at each use. State the Weierstrass approximation on the simplex together with the symmetrisation step as a lemma, since both the non-polynomial result and the LOCC proof rely on it.

Appendix G: LOCC feasibility SDP

1. Construction of the SDP

Let $|\psi\rangle$ be the initial pure state and ρ the target mixed state. By Ref. [50], $|\psi\rangle$ can be converted into ρ by LOCC if and only if there is a decomposition $\{p_i, |\psi_i\rangle\}$ of ρ such that

$$\sum_i p_i E_j(\psi_i) \le E_j(\psi) \qquad \text{for all } j = 1, \dots, d_A - 1. \tag{G1}$$

Since all these inequalities must hold for the same decomposition, testing them one monotone at a time is not sufficient. Instead, we combine the LP construction of Appendix F for all the Vidal monotones into a single SDP.

For each j, let $\mathcal{P}_j$ be the set of symmetric polynomials P_j of degree at most k that satisfy $P_j(\alpha^g) \le E_j(\alpha^g)$ on all grid points, and let

$$\epsilon_j(P_j) = \max_\alpha \left(P_j(\alpha) - E_j(\alpha)\right) \tag{G2}$$

be the maximal violation of P_j between grid points. If a decomposition $\{p_i, |\psi_i\rangle\}$ satisfies Eq. (G1), then its operator ω_k is feasible for Eq. (17) and, for every $P_j \in \mathcal{P}_j$,

$$\mathrm{tr}\left((O_{P_j} \otimes \mathbb{1}^B)\,\omega_k\right) = \sum_i p_i P_j(\alpha_i) \le \sum_i p_i E_j(\alpha_i) + \epsilon_j(P_j) \le E_j(\psi) + \epsilon_j(P_j), \tag{G3}$$

where α_i is the entanglement spectrum of $|\psi_i\rangle$. Therefore, if a certain decomposition can be transformed from $|\psi\rangle$ with LOCC, then it can violate the j'th inequality by at most the grid error $\epsilon_j(P_j)$ for each polynomial.

With the moments $\mu_\lambda(\Phi) = \mathrm{tr}\left((\frac{1}{f^\lambda}\Pi^\lambda \otimes \mathbb{1}^B)\Phi\right)$ of Appendix F, the optimization of Eq. (26) is computed by the SDP

$$\begin{aligned} s_k = \min_{\Omega_k,\, y^{(j)},\, s}\ & s \\ \text{s.t. } & \Omega_k \text{ feasible for Eq. (17)}, \quad y^{(j)} \ge 0 \ \ \forall j, \\ & \sum_g y_g^{(j)}\, s_\lambda(\alpha^g) = \mu_\lambda(\Omega_k) \quad \forall \lambda \vdash k,\ \forall j, \\ & \sum_g y_g^{(j)}\, E_j(\alpha^g) - E_j(\psi) \le s \quad \forall j. \end{aligned} \tag{G4}$$

For fixed Ω_k, the minimization over each $y^{(j)}$ is the dual LP of Appendix F with $F = E_j$, and therefore by strong duality

$$s_k = \min_{\Omega_k} \max_j \max_{P_j \in \mathcal{P}_j} \Big(\mathrm{tr}\left((O_{P_j} \otimes \mathbb{1}^B)\Omega_k\right) - E_j(\psi)\Big). \tag{G5}$$

Thus s_k is the smallest worst-case violation of Eq. (G1) over the relaxations of decompositions of ρ, where each violation is measured with the best grid-admissible polynomial. For a single monotone, $s_k = F_{\mathrm{LB}_k} - E_j(\psi)$ with $F = E_j$.

Let t be a probability vector with $t_j \geq 0$ and $\sum_j t_j = 1$. Then,

$$s_k = \min_{\Omega_k} \max_{t} \max_{P_j \in \mathcal{P}_j} \sum_j t_j \Big(\mathrm{tr}\left((O_{P_j} \otimes \mathbb{1}^B)\Omega_k\right) - E_j(\psi) \Big). \tag{G6}$$

With the substitution $Q_j = t_j P_j$, the constraint $P_j \in \mathcal{P}_j$ becomes $Q_j(\alpha^g) \leq t_j E_j(\alpha^g)$ for all g, and the objective is linear in Ω_k and in (t, Q_j). The feasible set of Ω_k is convex and compact, and the set of (t, Q_j) is convex. Again, here the minimum and the maximum can be exchanged by Ref. [53]:

$$\begin{aligned} s_k &= \min_{\Omega_k} \max_{t,\, Q_j} \sum_j \Big(\mathrm{tr}\left((O_{Q_j} \otimes \mathbb{1}^B)\Omega_k\right) - t_j E_j(\psi) \Big) \\ &= \max_{t,\, Q_j} \min_{\Omega_k} \sum_j \Big(\mathrm{tr}\left((O_{Q_j} \otimes \mathbb{1}^B)\Omega_k\right) - t_j E_j(\psi) \Big). \end{aligned} \tag{G7}$$

Let t^* and Q_j^* be the optimal vector t and optimal polynomials, which are given by the dual variables of the SDP (G4); for $t_j^* > 0$, $Q_j^* = t_j^* P_j^*$ with $P_j^* \in \mathcal{P}_j$. By Eq. (G7), they satisfy

$$\sum_j \Big(\mathrm{tr}\left((O_{Q_j^*} \otimes \mathbb{1}^B)\Omega_k\right) - t_j^* E_j(\psi) \Big) \geq s_k \tag{G8}$$

for every feasible Ω_k.

Let $\epsilon_j = \max_\alpha \left(Q_j^*(\alpha) - t_j^* E_j(\alpha)\right)$, which equals $t_j^* \, \epsilon_j(P_j^*)$ for $t_j^* > 0$. If $|\psi\rangle$ can be converted into ρ by LOCC, some decomposition satisfies Eq. (G1), and its ω_k is feasible. As in Eq. (G3), $\mathrm{tr}\left((O_{Q_j^*} \otimes \mathbb{1}^B)\,\omega_k\right) = \sum_i p_i Q_j^*(\alpha_i) \leq t_j^* E_j(\psi) + \epsilon_j$, and therefore Eq. (G8) gives

$$s_k \leq \sum_j \Big(\mathrm{tr}\left((O_{Q_j^*} \otimes \mathbb{1}^B)\,\omega_k\right) - t_j^* E_j(\psi) \Big) \leq \sum_j \epsilon_j. \tag{G9}$$

Thus, if $s_k > \sum_j \epsilon_j$, $|\psi\rangle$ cannot be converted into ρ by LOCC. If the condition does not hold, the test does not tell us anything, in which case, for $k \geq 3$, we can extract the decomposition from Ω_k and repair it as in Sec. VII, and if the criterion Eq. (G1) holds, the state can be converted into ρ by LOCC. In this procedure, the grid errors ϵ_j need to be computed in a postprocessing step from the polynomials returned by the solver.

2. Detection at finite level

We impose $P_j \leq E_j$ on all spectra. Consider the decomposition $\{p_i, |\varphi_i\rangle\langle\varphi_i|\}$ of ρ that minimizes

$$s^* = \max_t \sum_j t_j \left(\left(\sum_i p_i E_j(\varphi_i) \right) - E_j(\psi) \right) \tag{G10}$$

over probability vectors t. If $|\psi\rangle \overset{\mathrm{LOCC}}{\nrightarrow} \rho$, at least one Vidal monotone E_j will be exceeded by the decomposition and setting $t_j = 1$, we get that $s^* > 0$. We know $\widetilde{s}_k \leq s^*$ since we are relaxing the set of feasible ω_k. By App. A, as $k \to \infty$, the few-copy marginals of Ω_k approach mixtures $\sum q_i \, |\phi_i\rangle\langle\phi_i|^{\otimes n}$ with $\{q_i, |\phi_i\rangle\langle\phi_i|\}$ a decomposition of ρ, and since by App. F 2 each E_j can be approximated to arbitrary precision from below, $\widetilde{s}_k \to s^*$ as $k \to \infty$ and every non-convertible pair is eventually detected.

[41] Deeper proof. Tighten this proof. (i) $\widetilde{s}_k$ is never defined: define it as s_k with $P_j \leq E_j$ imposed on all spectra. (ii) A decomposition minimising s^* exists because decompositions with at most r^2 terms form a compact set and the objective is continuous; say so. (iii) The convergence $\widetilde{s}_k \to s^*$ needs the polynomial approximations to be uniform in j and the de Finetti step to hold for all j at once; write the η/δ argument as in App. F 2.

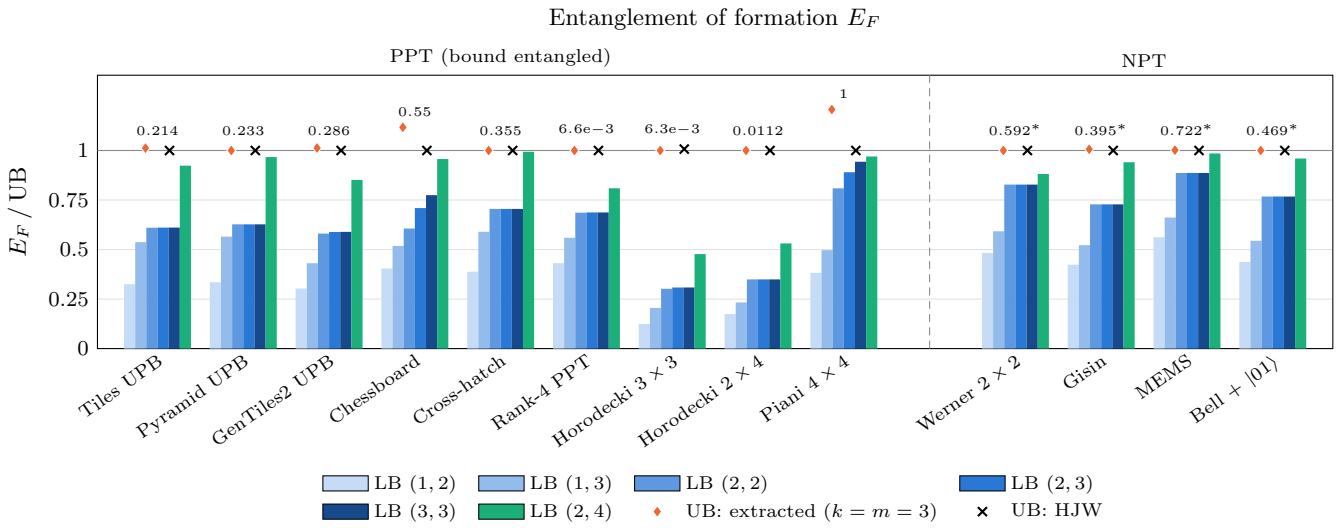


FIG. 5. Lower bounds on the entanglement of formation E_F for different PPT and NPT states, from the SDP with embedded LP. Bars: lower bounds from the simplified hierarchy at the levels (k, m) shown in the legend. Markers: upper bounds from the decomposition extracted from the SDP solution at $k = m = 3$ (orange diamonds) and from HJW gradient descent over decompositions (black crosses). All values are divided by the best upper bound UB, printed above each state. Here, UB is the exact convex roof for two qubits (marked *), and the lowest of the two upper bounds otherwise. Notice that the gap between upper and lower bounds decreases with the hierarchy level.

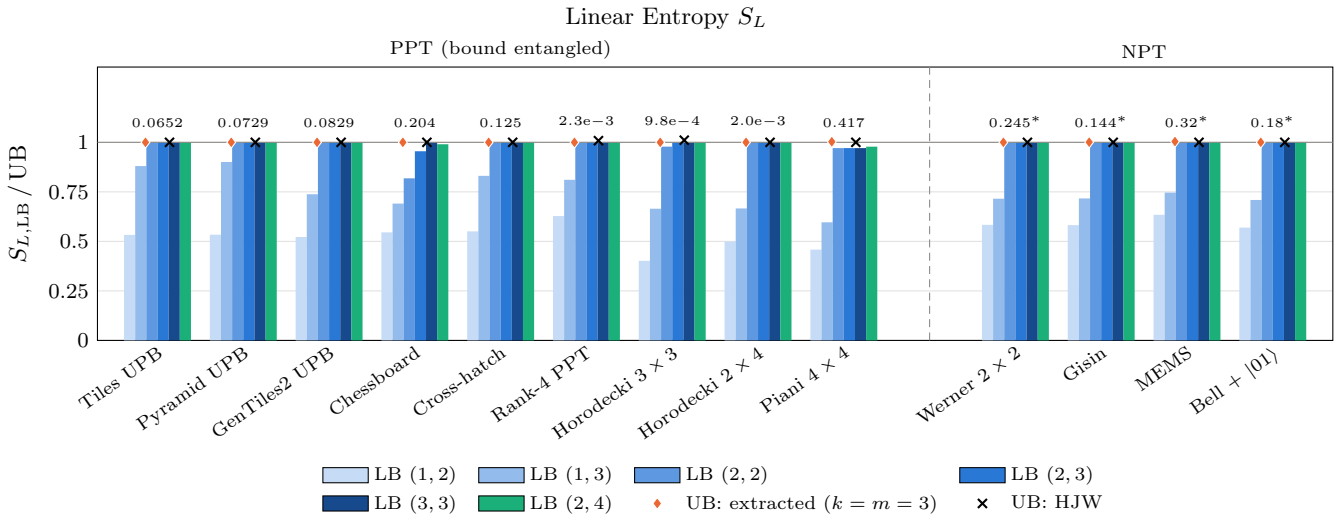


FIG. 6. Lower bounds on the linear entropy for different PPT and NPT states. Bars: lower bounds from the simplified hierarchy at the levels (k, m) shown in the legend. Markers: upper bounds from the decomposition extracted from the SDP solution at $k = m = 3$ (orange diamonds) and from HJW gradient descent over decompositions (black crosses). All values are divided by the best upper bound UB, printed above each state. Here, UB is the exact convex roof for two qubits (marked *), and the lowest of the two upper bounds otherwise.

Appendix H: Further analysis

In this appendix, we provide additional numerical results for the entanglement of formation (Fig. 5), the linear entropy (Fig. 6) and the Vidal monotones (Fig. 7). We also show how the LP adapts to different states by choosing different lower-bounding polynomials, which are tangent to the reference function at different points of the spectrum (Fig. 8).

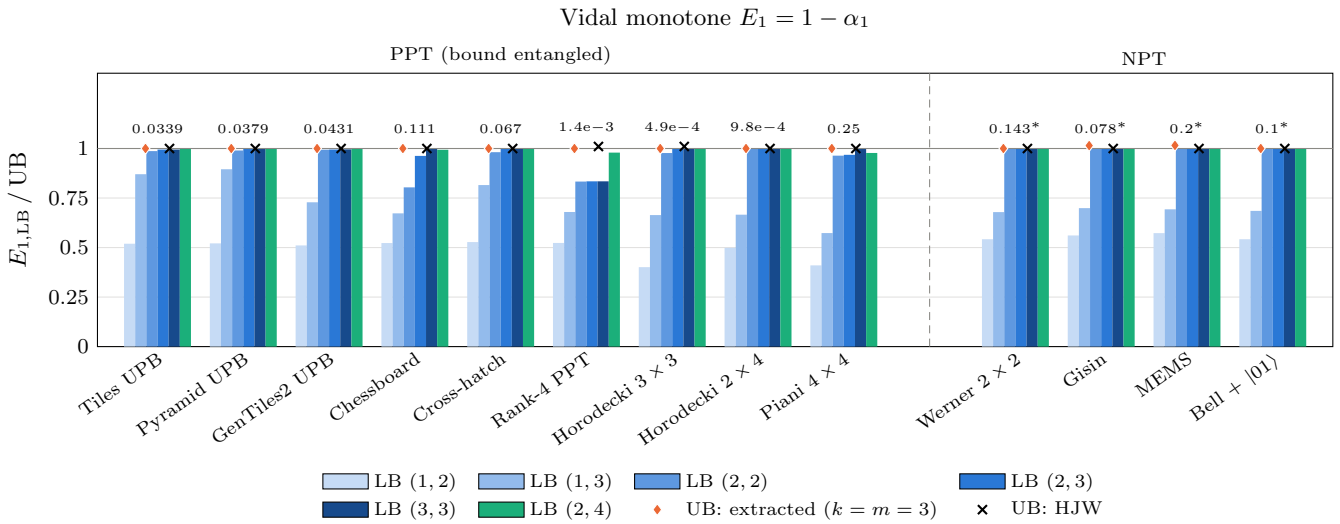


FIG. 7. Lower bounds on the Vidal monotone E_1 for different PPT and NPT states, from the SDP with embedded LP. Bars: lower bounds from the simplified hierarchy at the levels (k, m) shown in the legend. Markers: upper bounds from the decomposition extracted from the SDP solution at $k = m = 3$ (orange diamonds) and from HJW gradient descent over decompositions (black crosses). All values are divided by the best upper bound UB, printed above each state. Here, UB is the exact convex roof for two qubits (marked *), and the lowest of the two upper bounds otherwise.

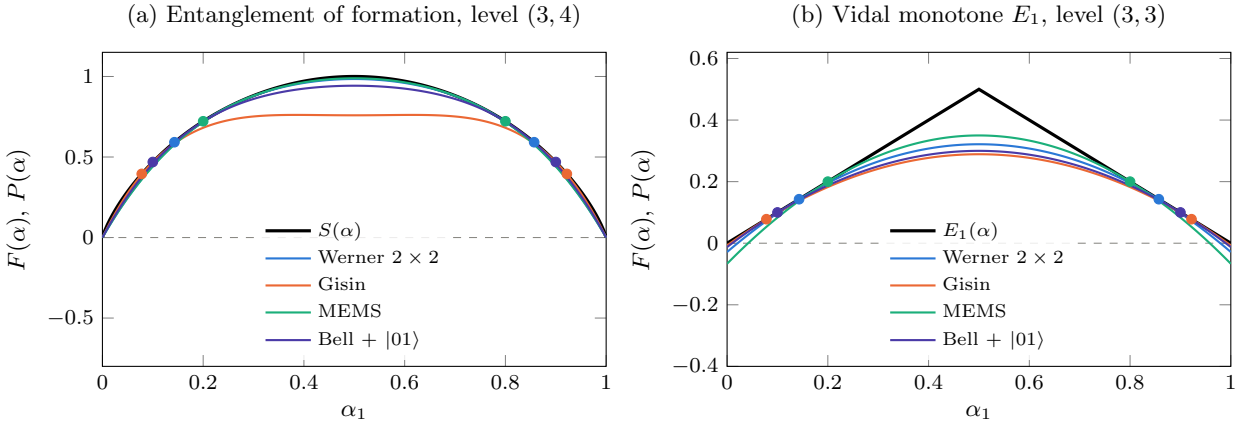


FIG. 8. Polynomials $P \leq F$ extracted from the SDP with embedded LP for different 2-qubit states. For each state, this is the polynomial that yields the best bound on the convex roof on the grid, at level $(3, 4)$ for E_F (a) and at level $(3, 3)$ for E_1 (b). Here we show that the LP is adaptive depending on the state and there is not a single polynomial that yields the best bound on the convex roof for all states at the same level. The states used are { Werner 2×2: singlet weight $p = 0.8$; Gisin: $\lambda = 0.9$, $\theta = \pi/8$; MEMS: $C = 0.8$; Bell + $|01\rangle$: $p = 0.6$ }. The polynomials are shown as a function of the entanglement spectrum $\alpha = (\alpha_1, \alpha_2) = (\alpha_1, 1 - \alpha_1)$.

Appendix I: Numerical implementation

All the SDPs are solved with the interior-point conic solver QICS [54], with its default tolerances. The PPT entangled states from the figures come from the library `bound-entangled` [55] and the NPT ones are built with `toqito` [56]. For the Horodecki 3×3 and 2×4 states we take the parameter $a = 0.5$ in Figs. 5, 6 and 7. The parameters of the NPT states are the ones given in the caption of Fig. 8.

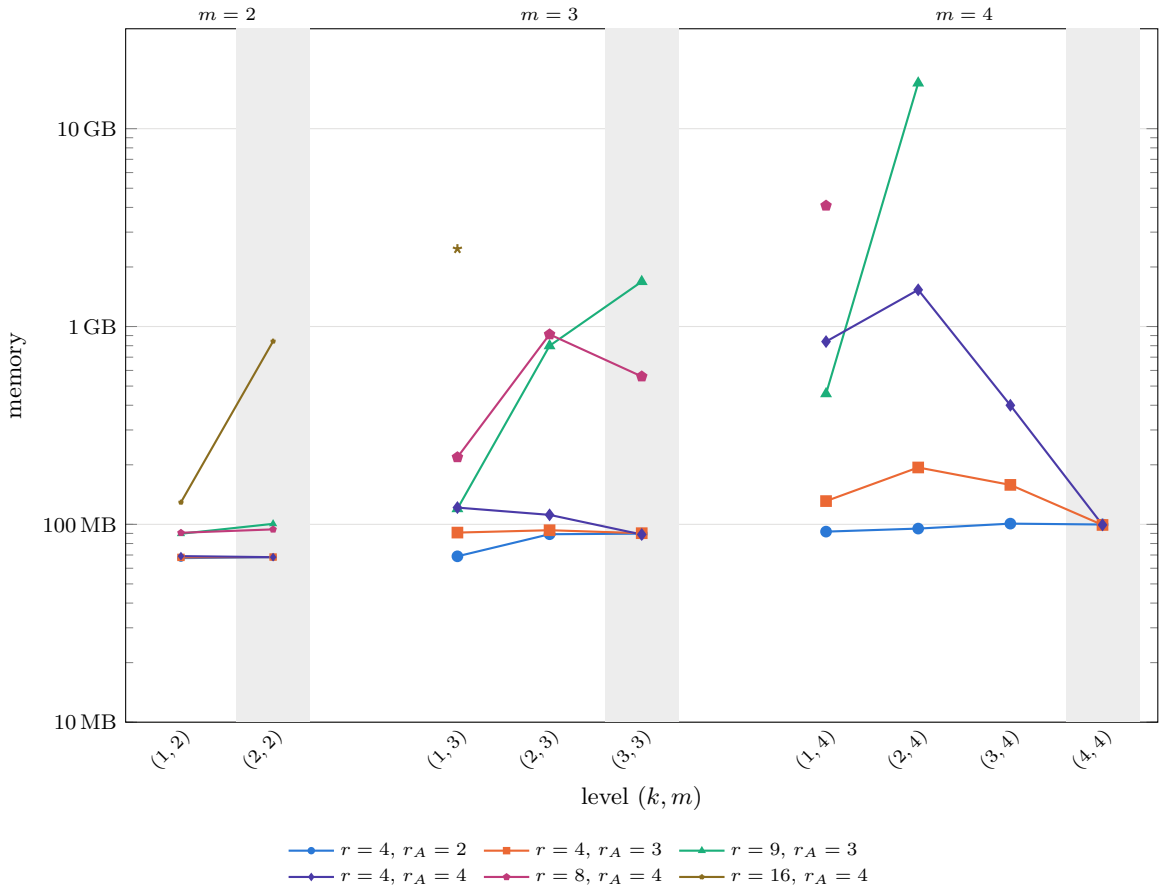


FIG. 9. Memory needed by the SDP (19) with the QICS solver [54] at different levels for random states ρ with $r = \operatorname{rank}(\rho)$ and $r_A = \operatorname{rank}(\operatorname{tr}_B(\rho))$. When $k = m$, the full SDP (17) is used.

1. Upper bounds by gradient descent

By the HJW theorem [48], every decomposition of $\rho = \sum_i \lambda_i |e_i\rangle\langle e_i|$ into M pure states is obtained by mixing the vectors $\sqrt{\lambda_i}\,|e_i\rangle$ with an $M \times M$ unitary. Therefore every unitary provides a decomposition of ρ and hence an upper bound on the convex roof. As in Ref. [57], we minimize this upper bound by gradient descent over the unitaries, with $M = r^2$ and several random starting points, and keep the best value.

[42] **New content.** Implementation chapter: code structure, how a state goes through the pipeline, solver choice (picos first, then a native sparse QICS model using about 3 GB instead of 15 GB for $d = 9$, $k = 3$, matching picos to 8 digits), the memory model against measurements (33 measured cases), the numerical pitfalls (QICS cones caching their structure, the normalised rank-reduction objective, HiGHS tolerances tightened to 10^{-10}, the missing index interleaving in the first tensor reshape, the 2 h limit on background jobs and resumable result files), reproducibility (where code and data live, how each figure is regenerated) and run times.